Trees

Alexander Golovnev

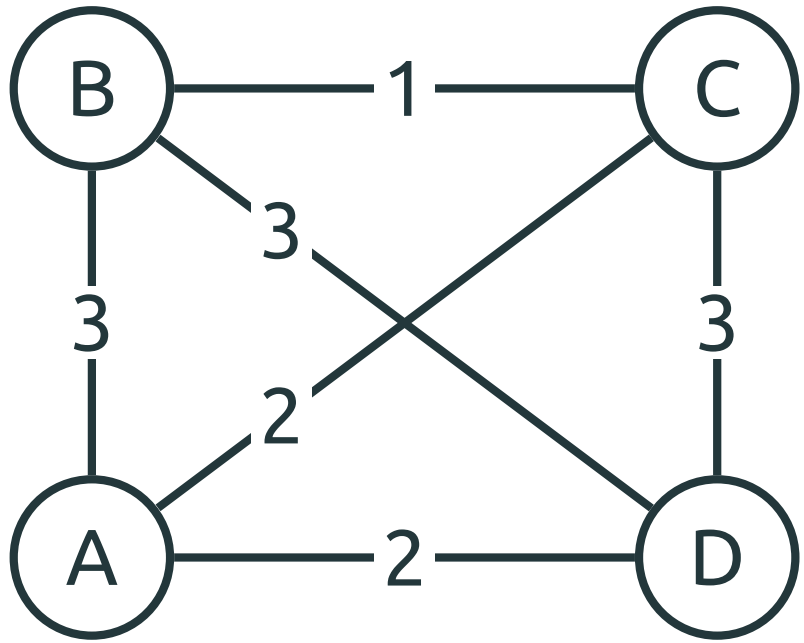
Outline

Road Repair

Trees

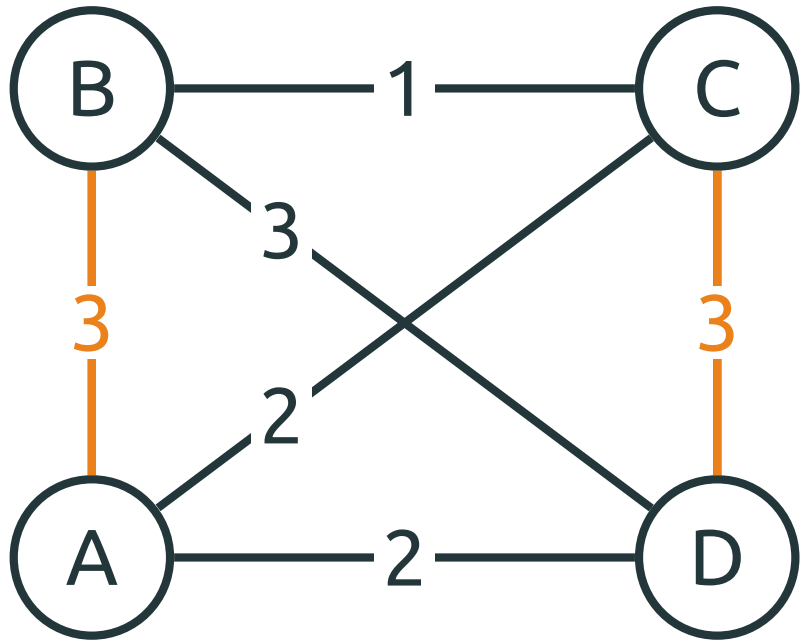
Minimum Spanning Tree

Road Repair



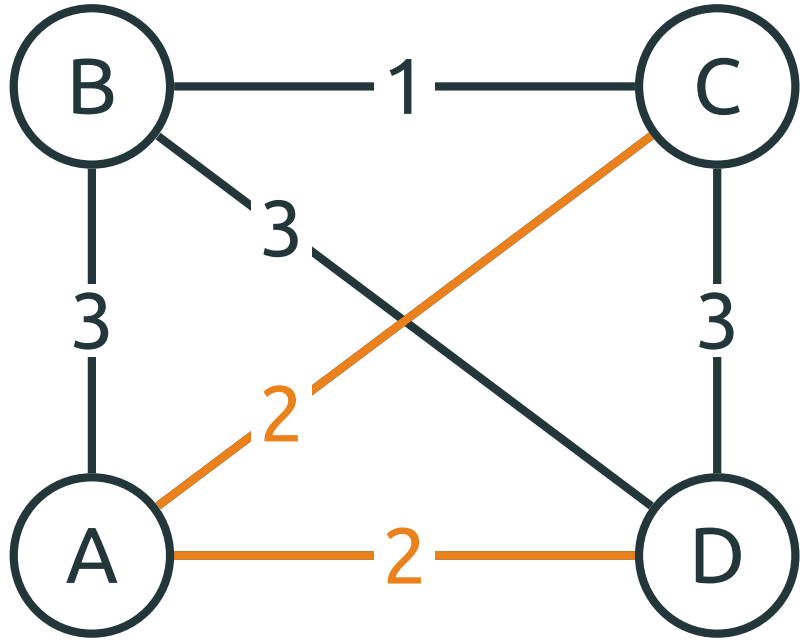
Road Repair

No pair of edges can connect all cities



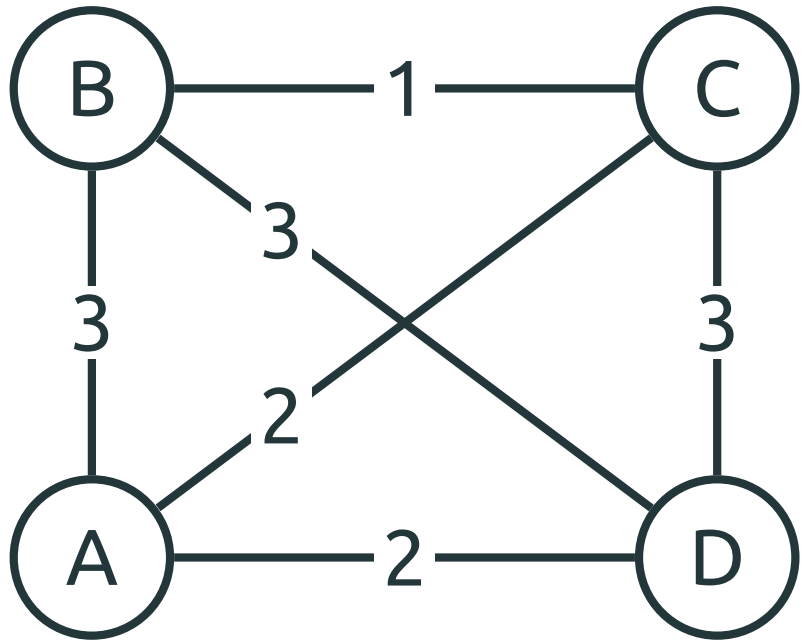
Road Repair

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Road Repair

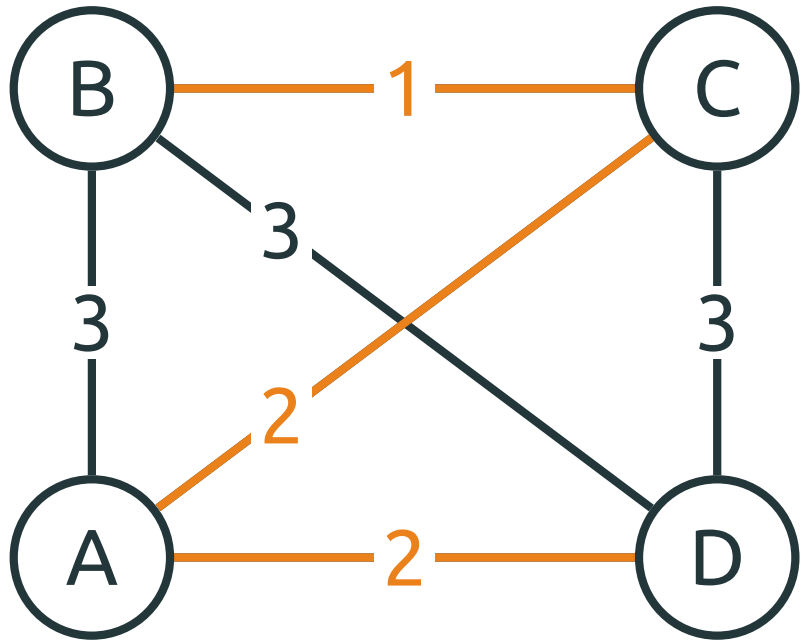
We need at least three edges



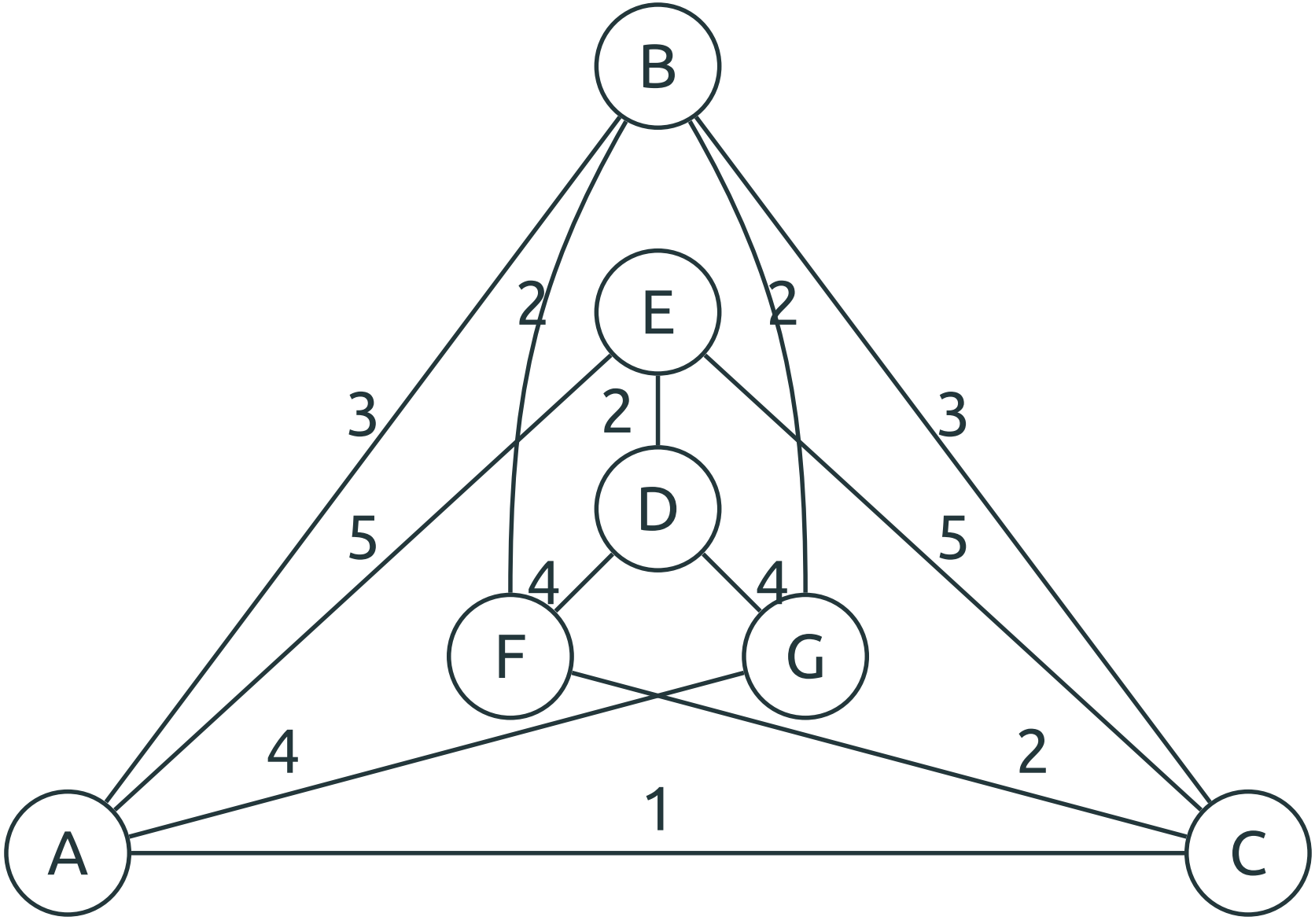
Road Repair

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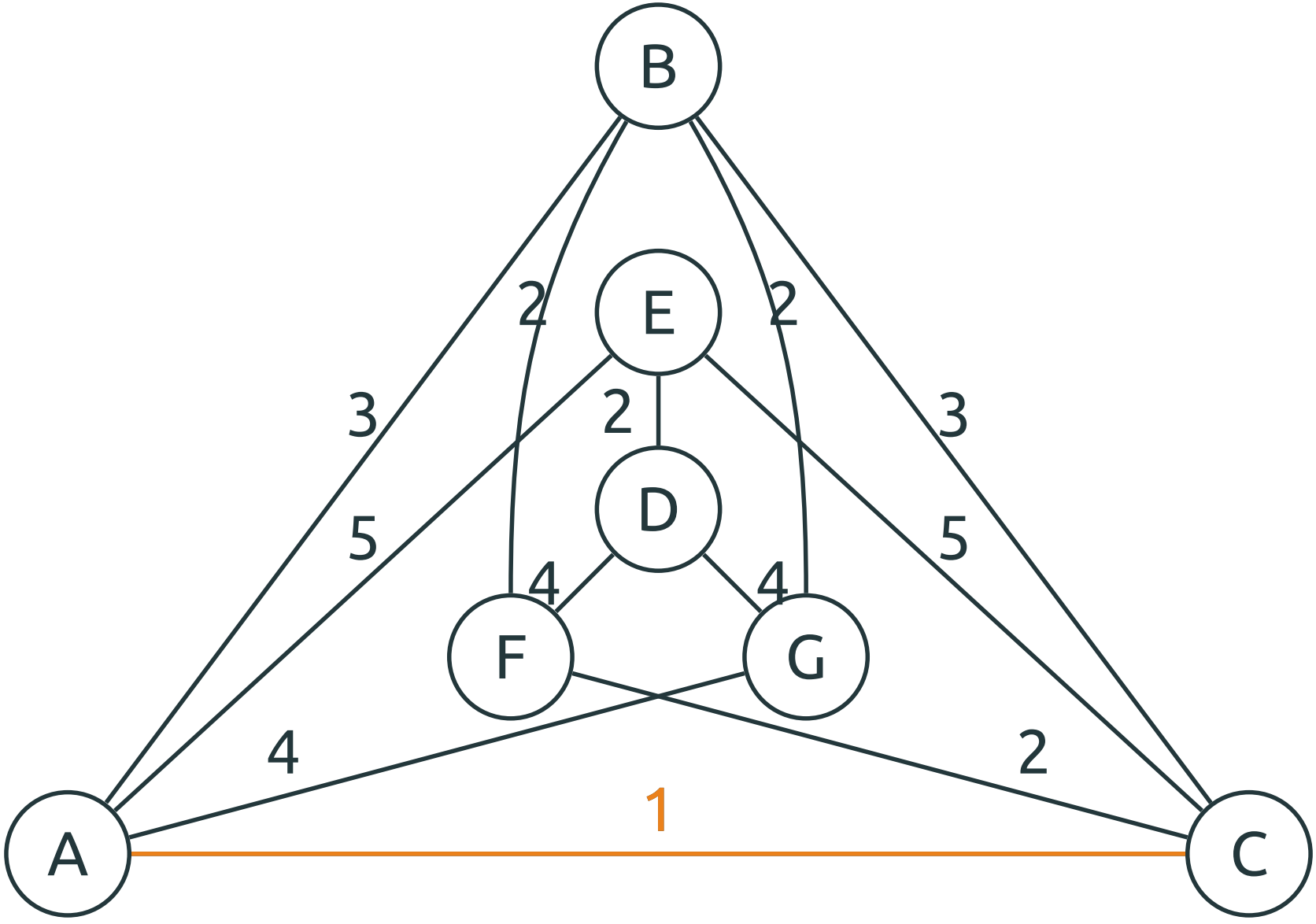
Three shortest edges work!



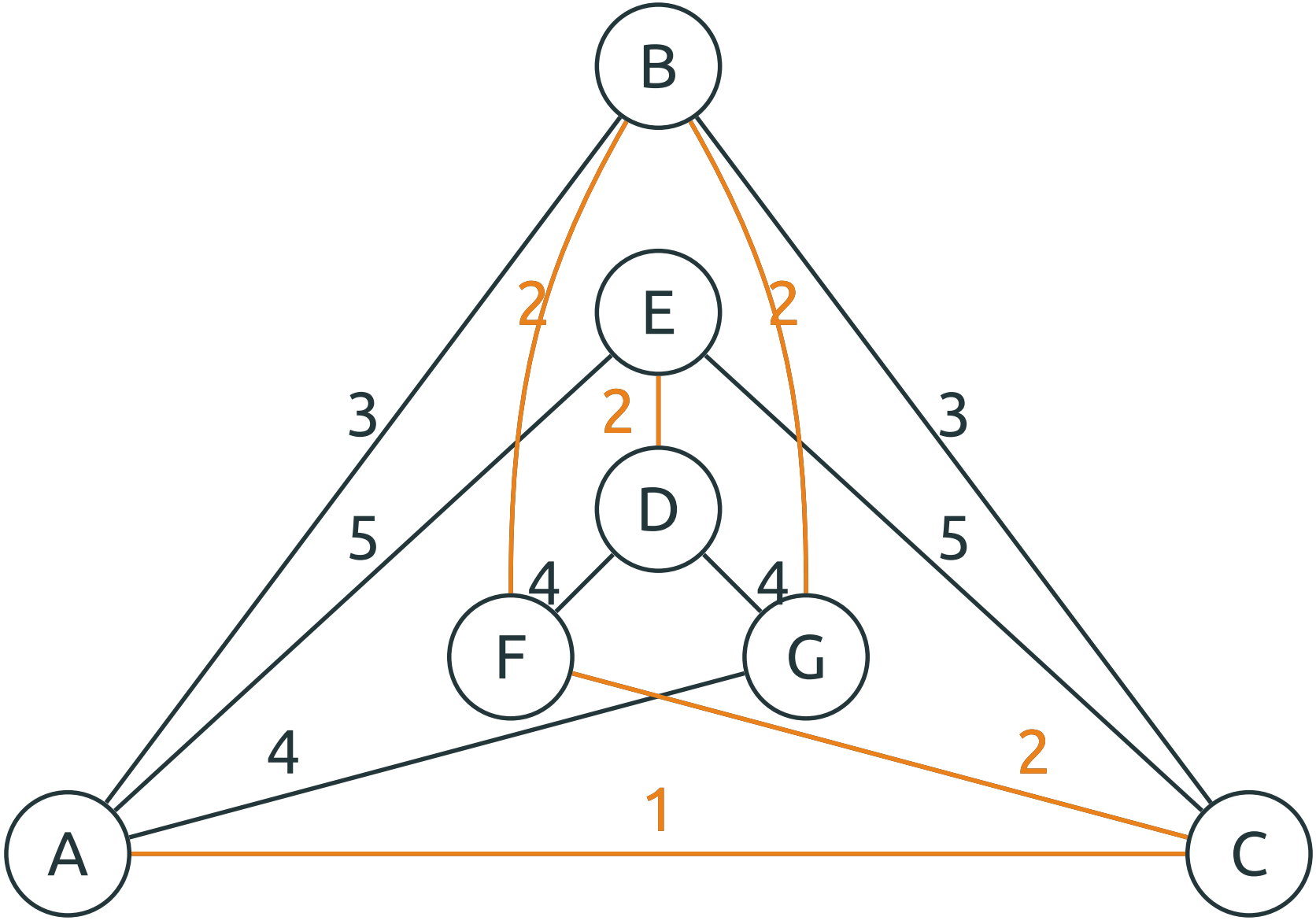
Road Repair



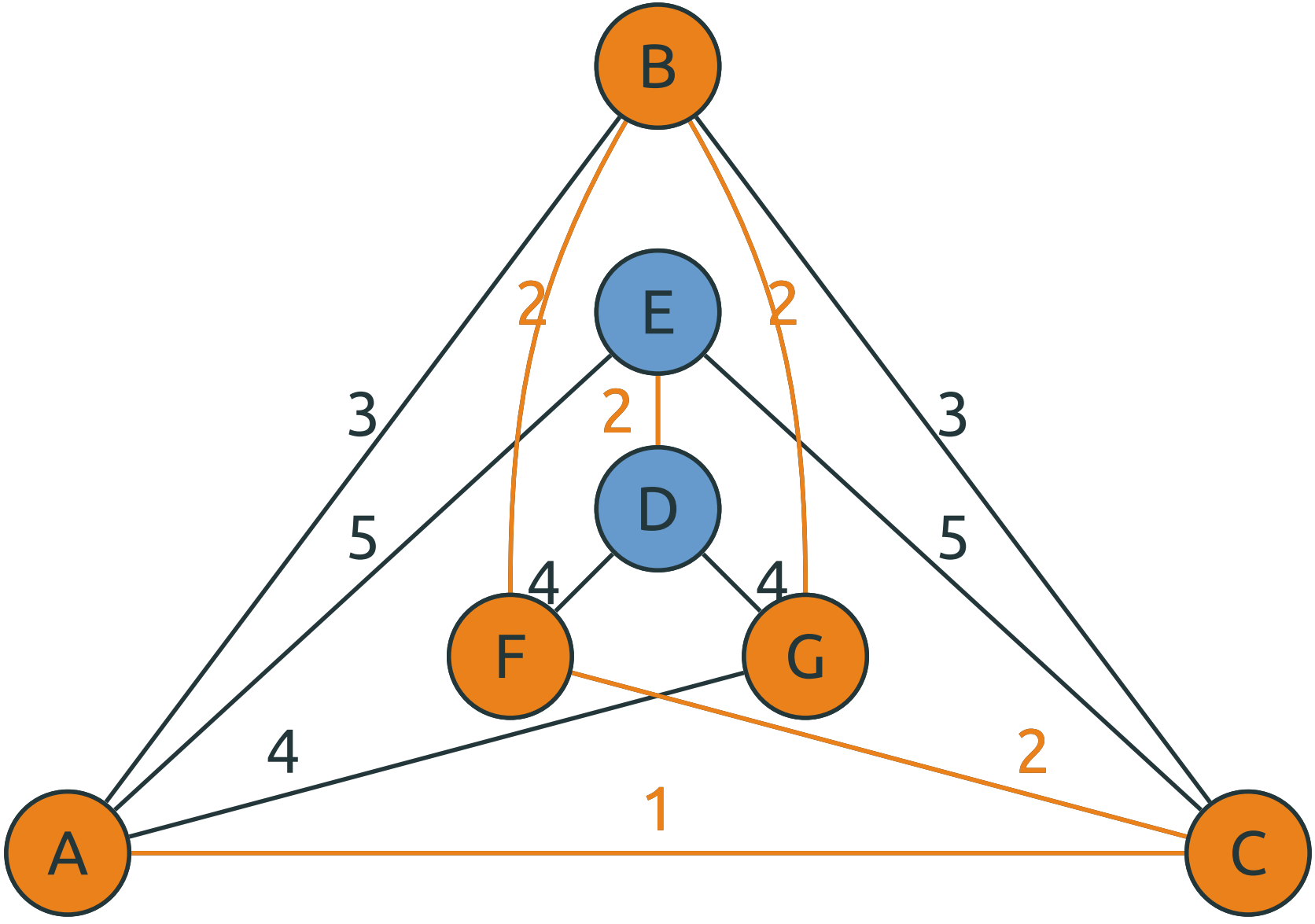
Road Repair



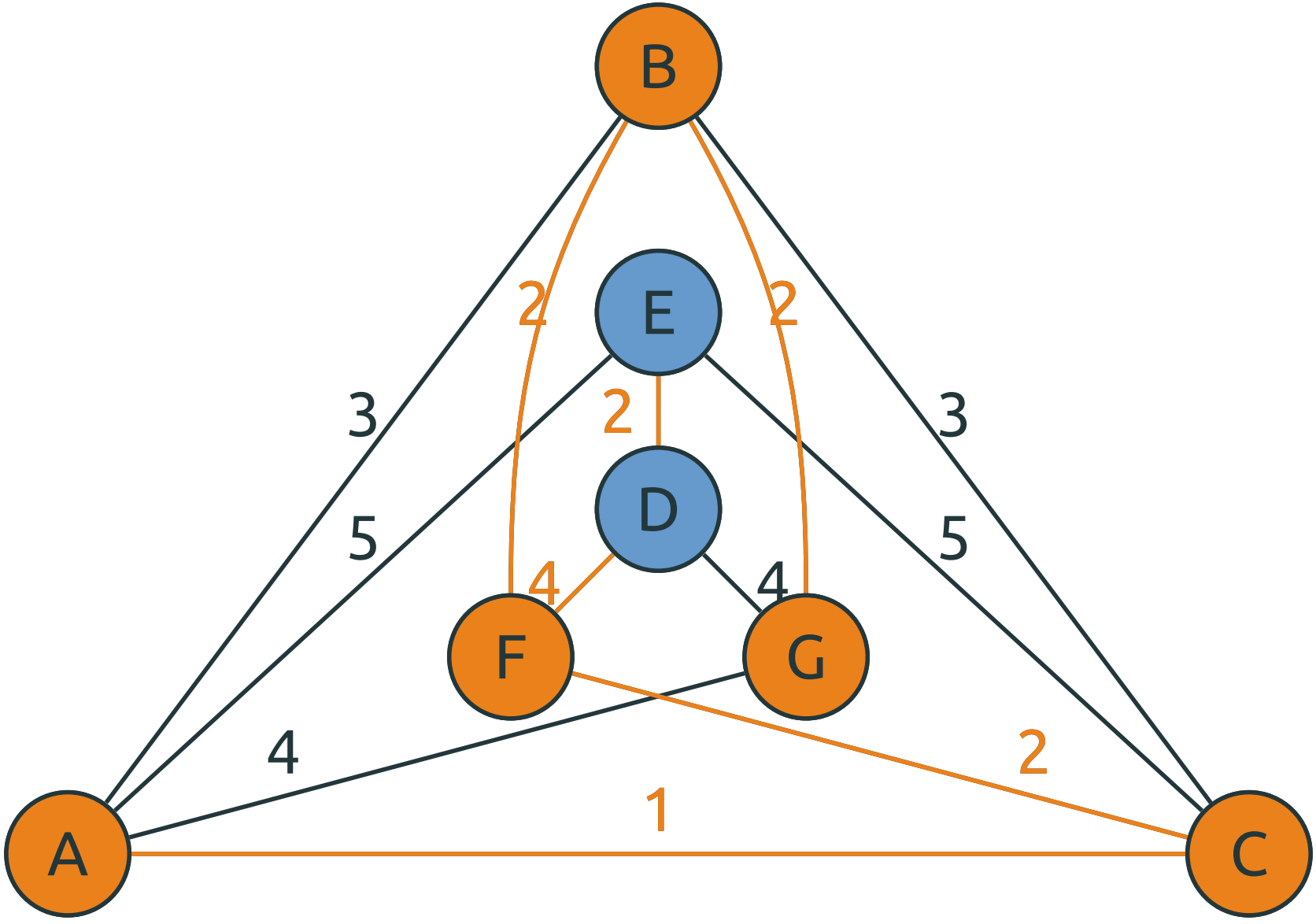
Road Repair



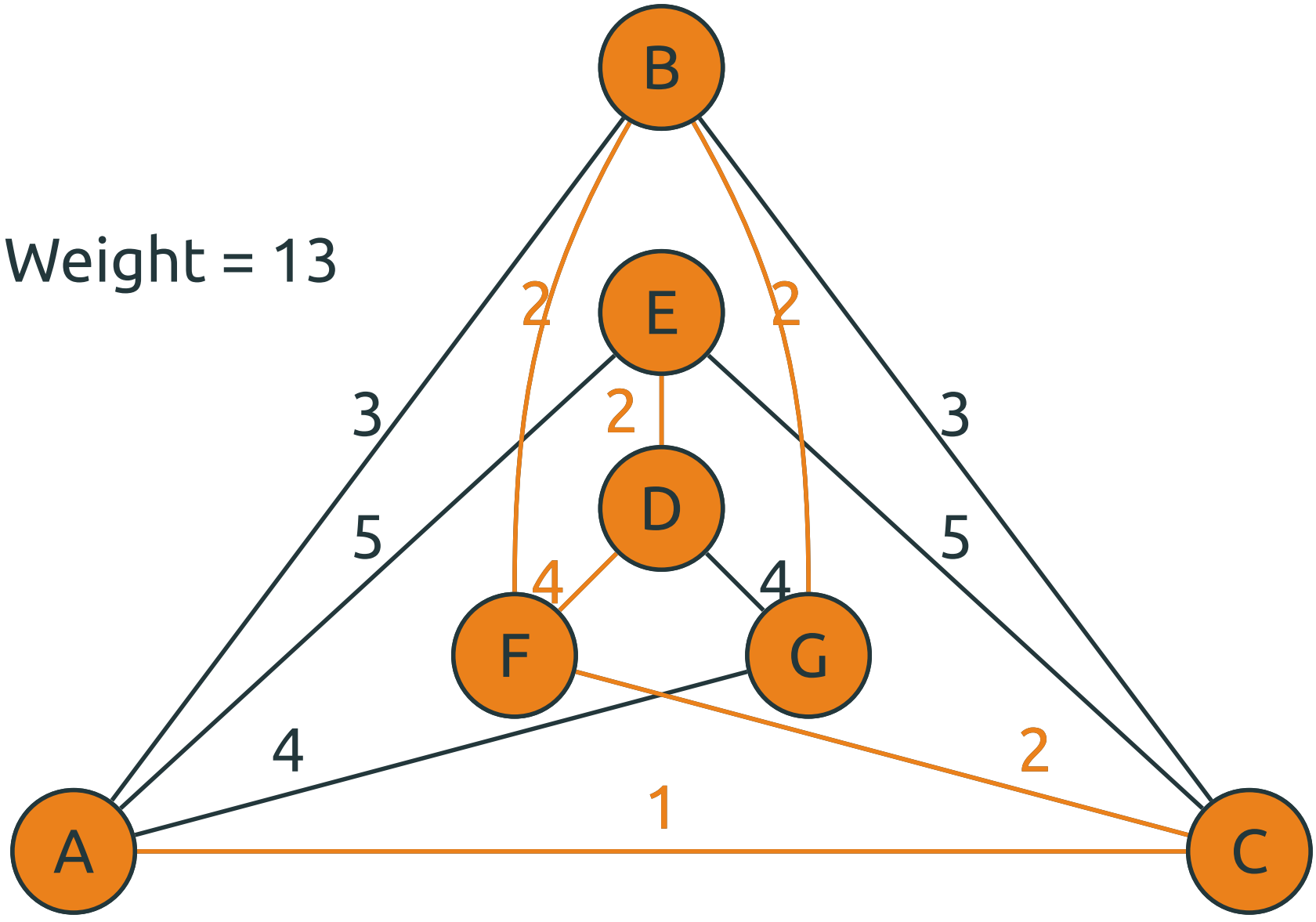
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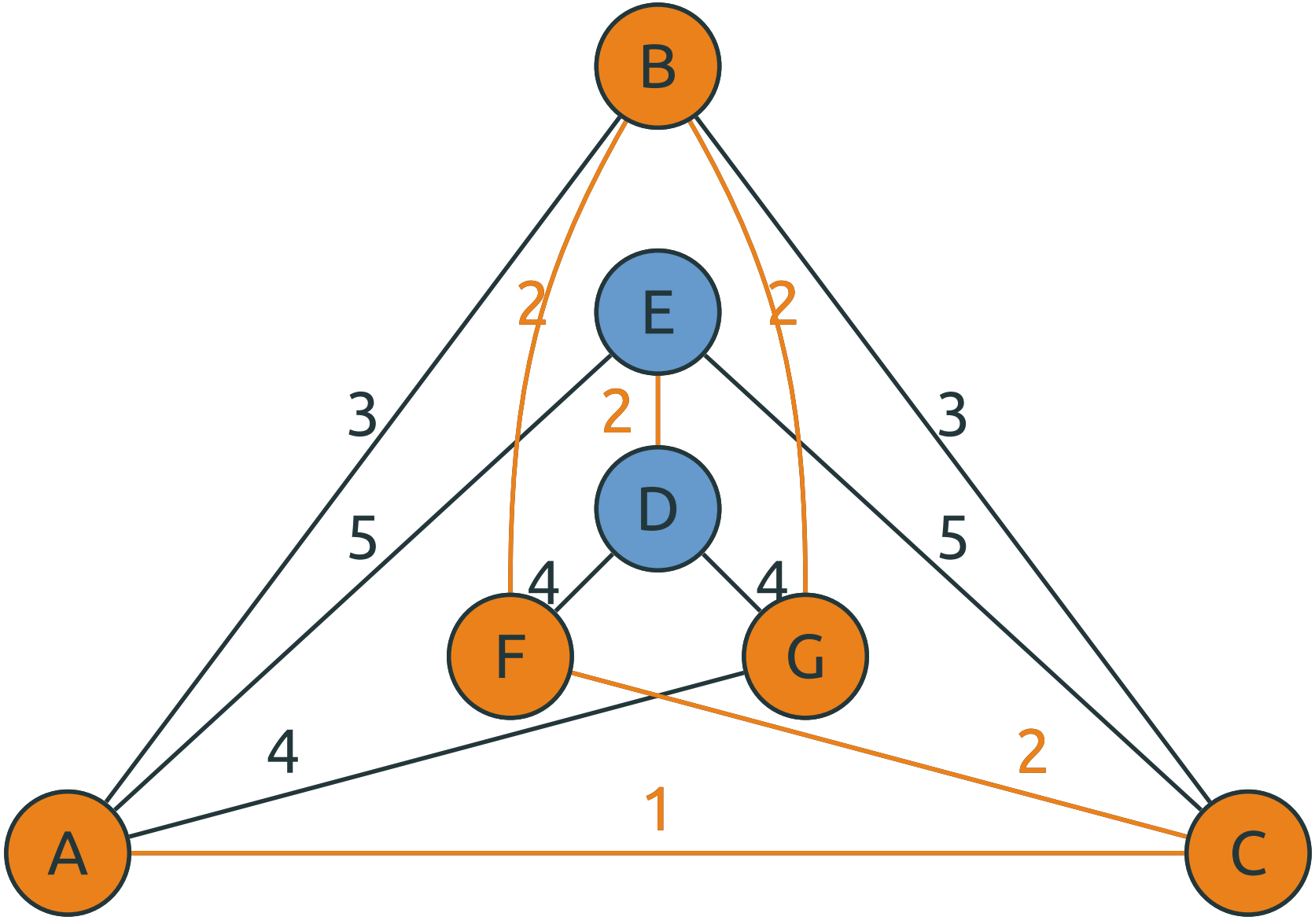
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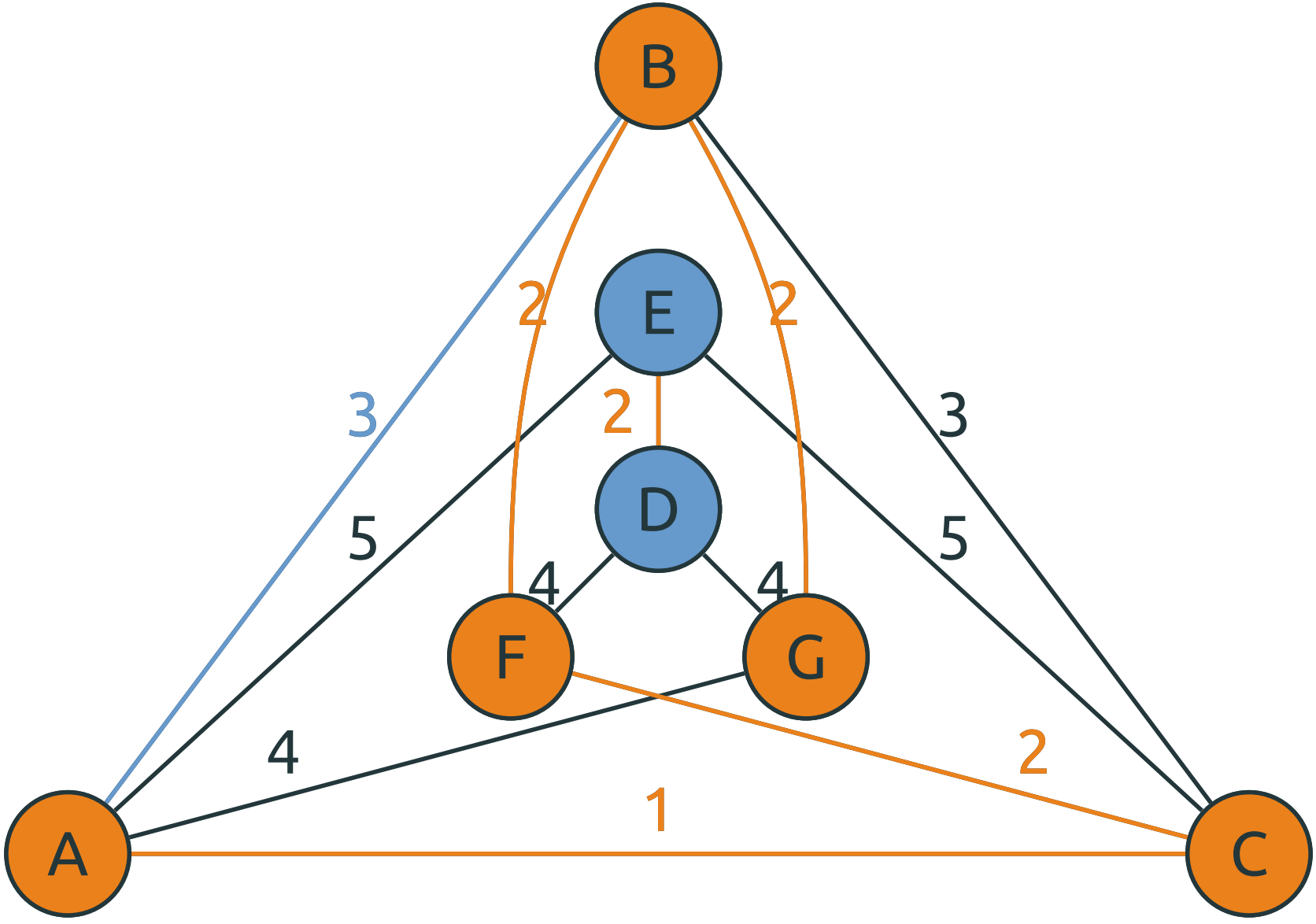
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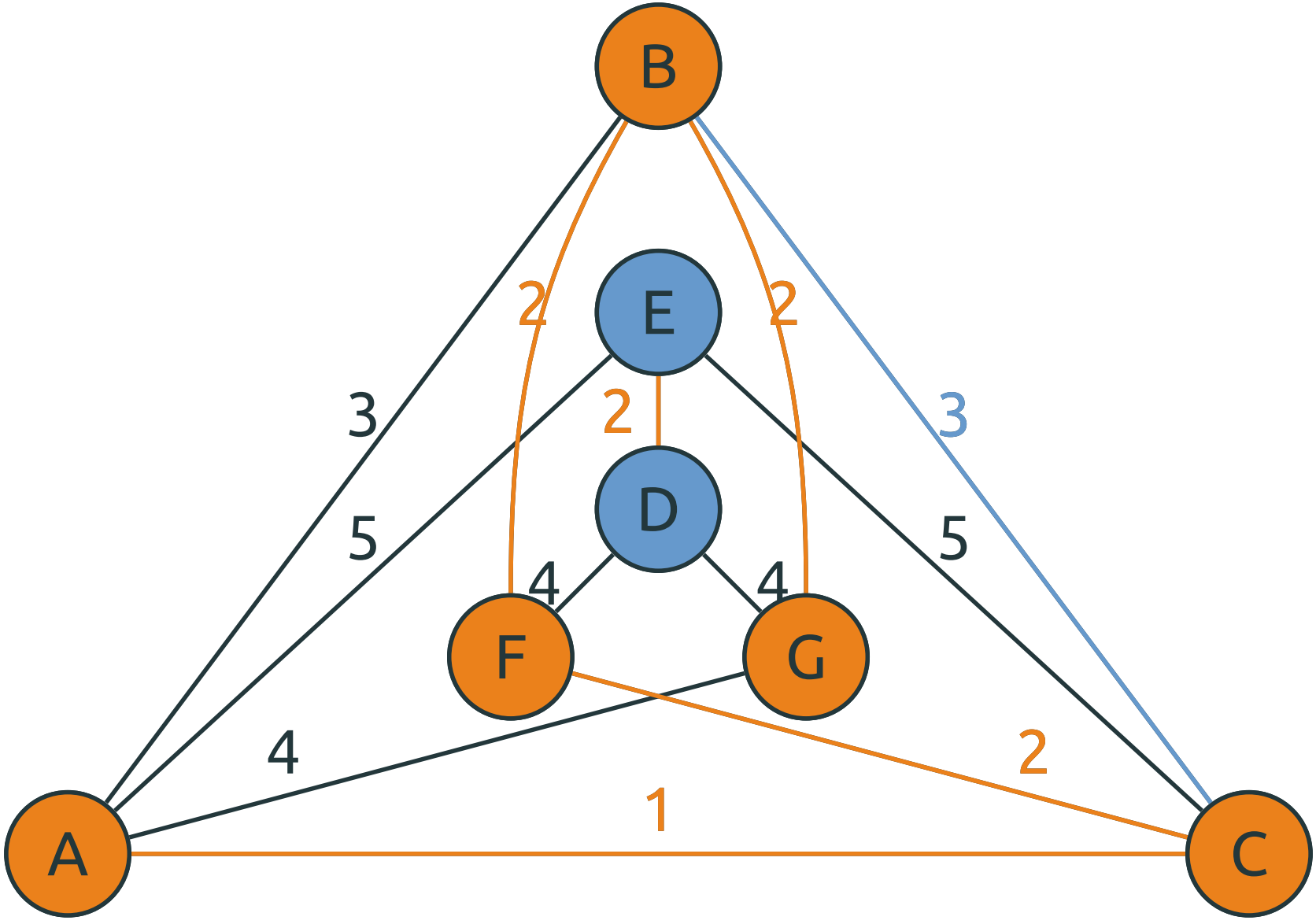
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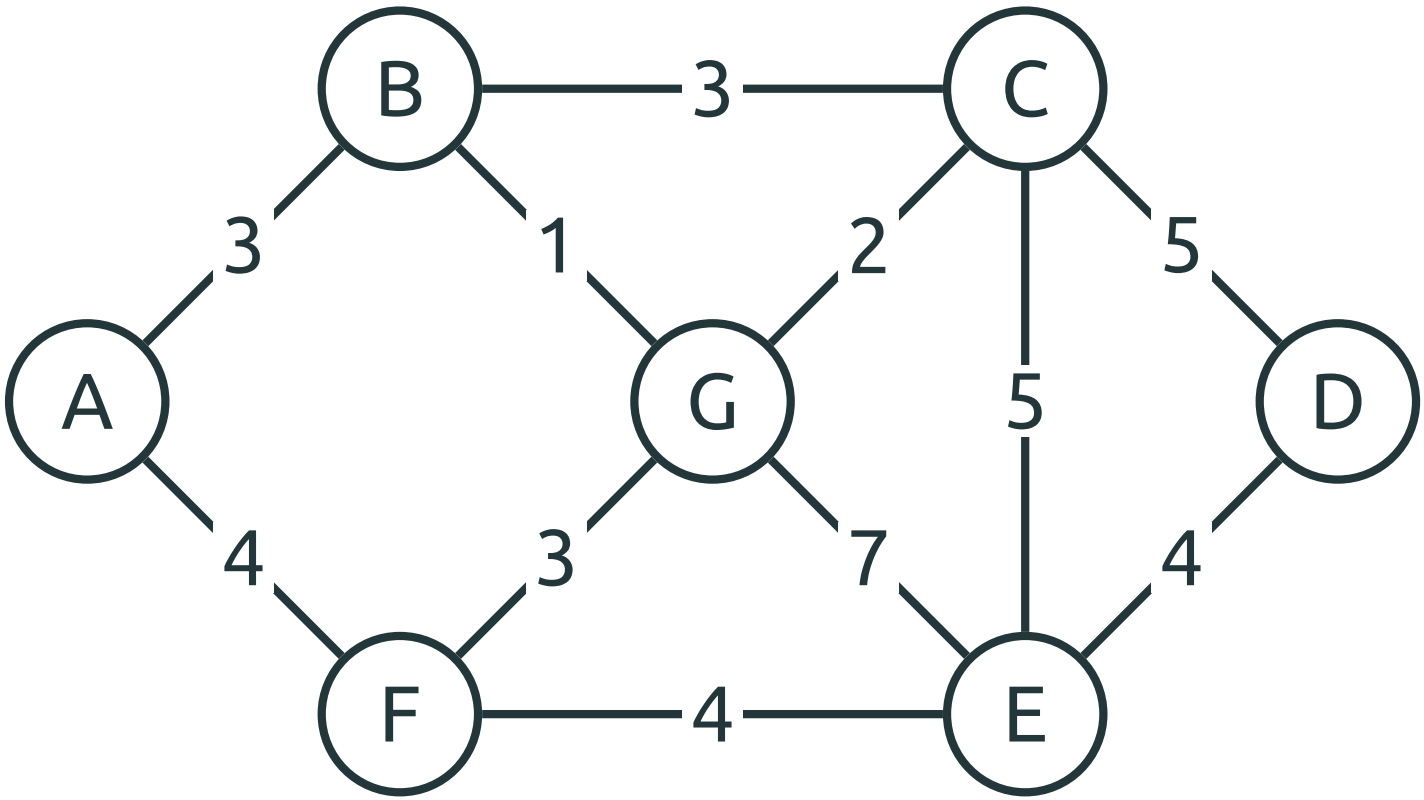
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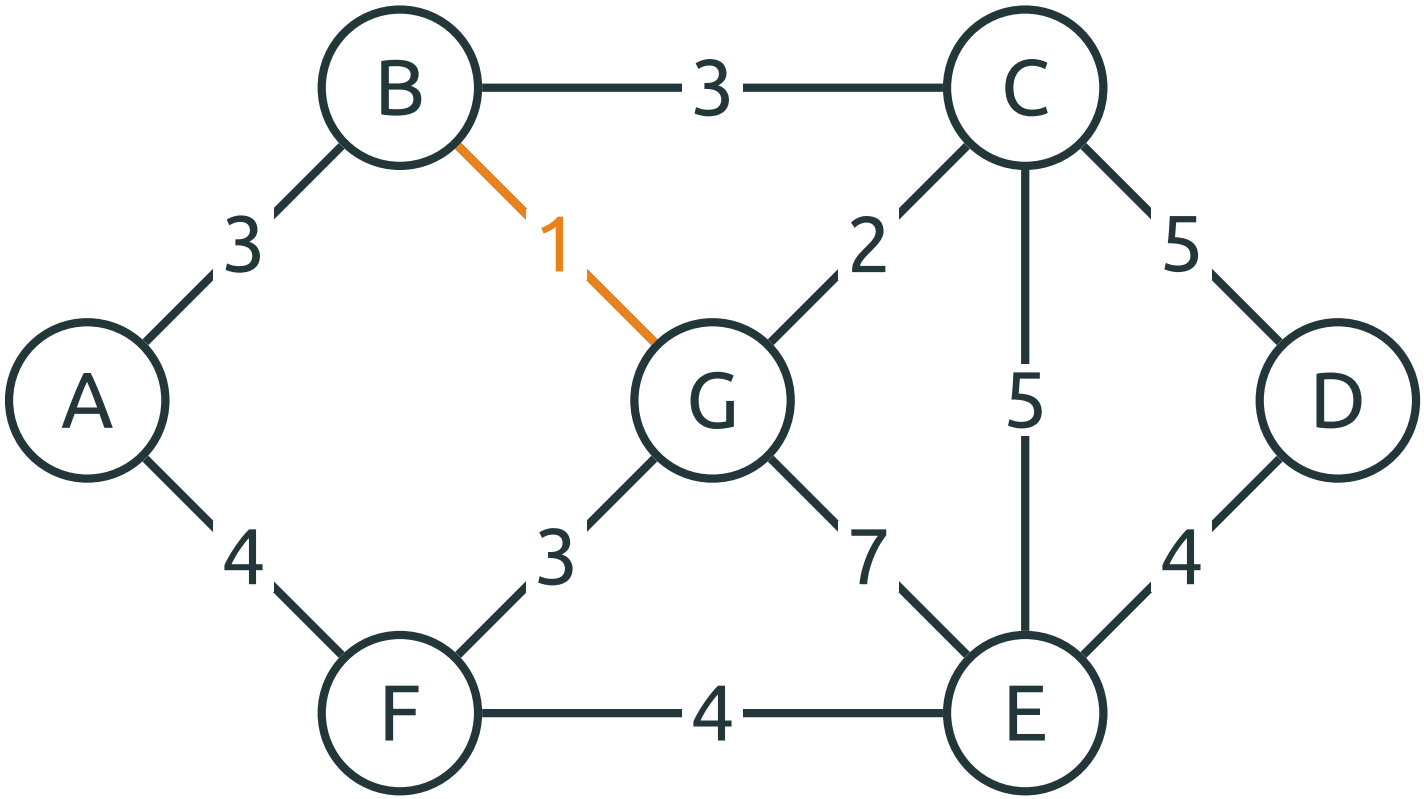
Road Repair



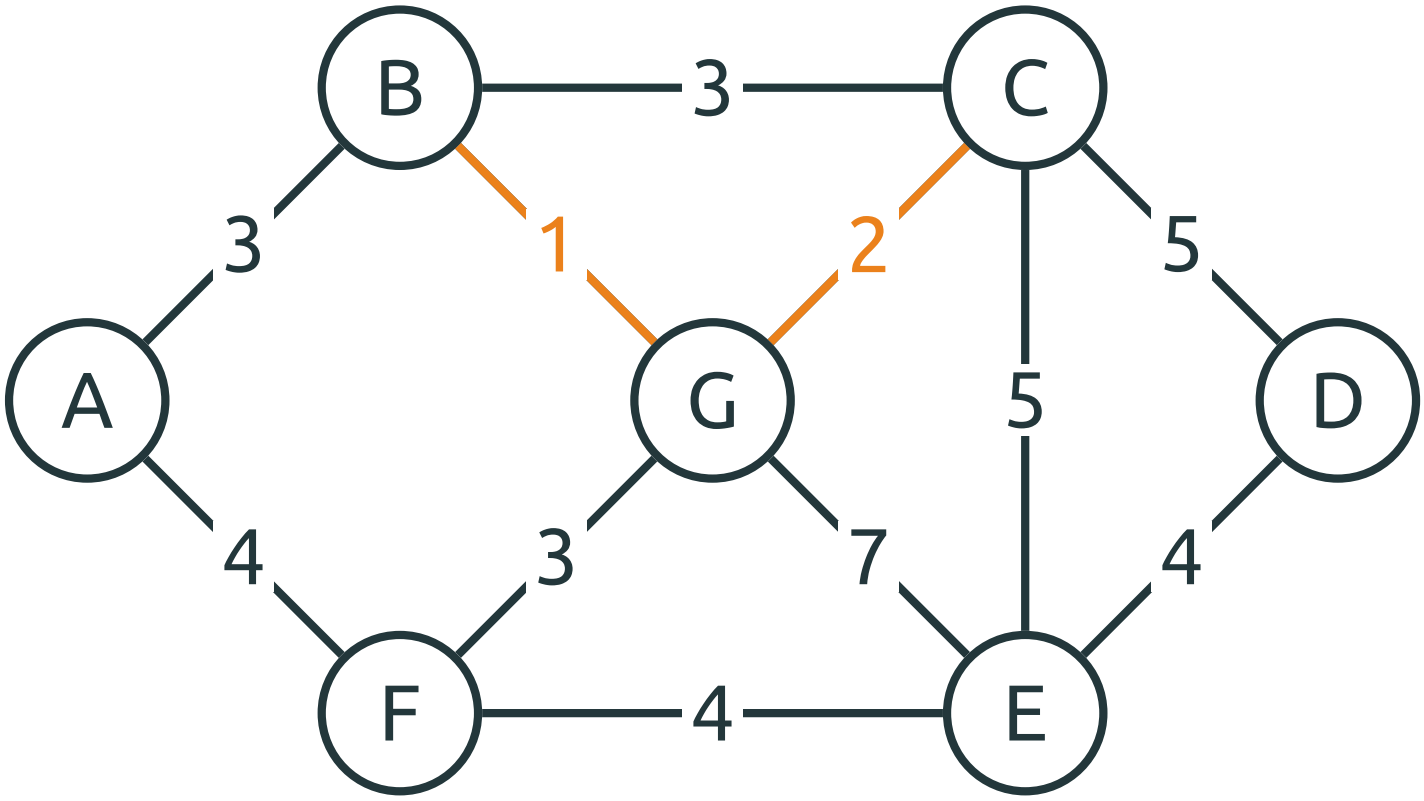
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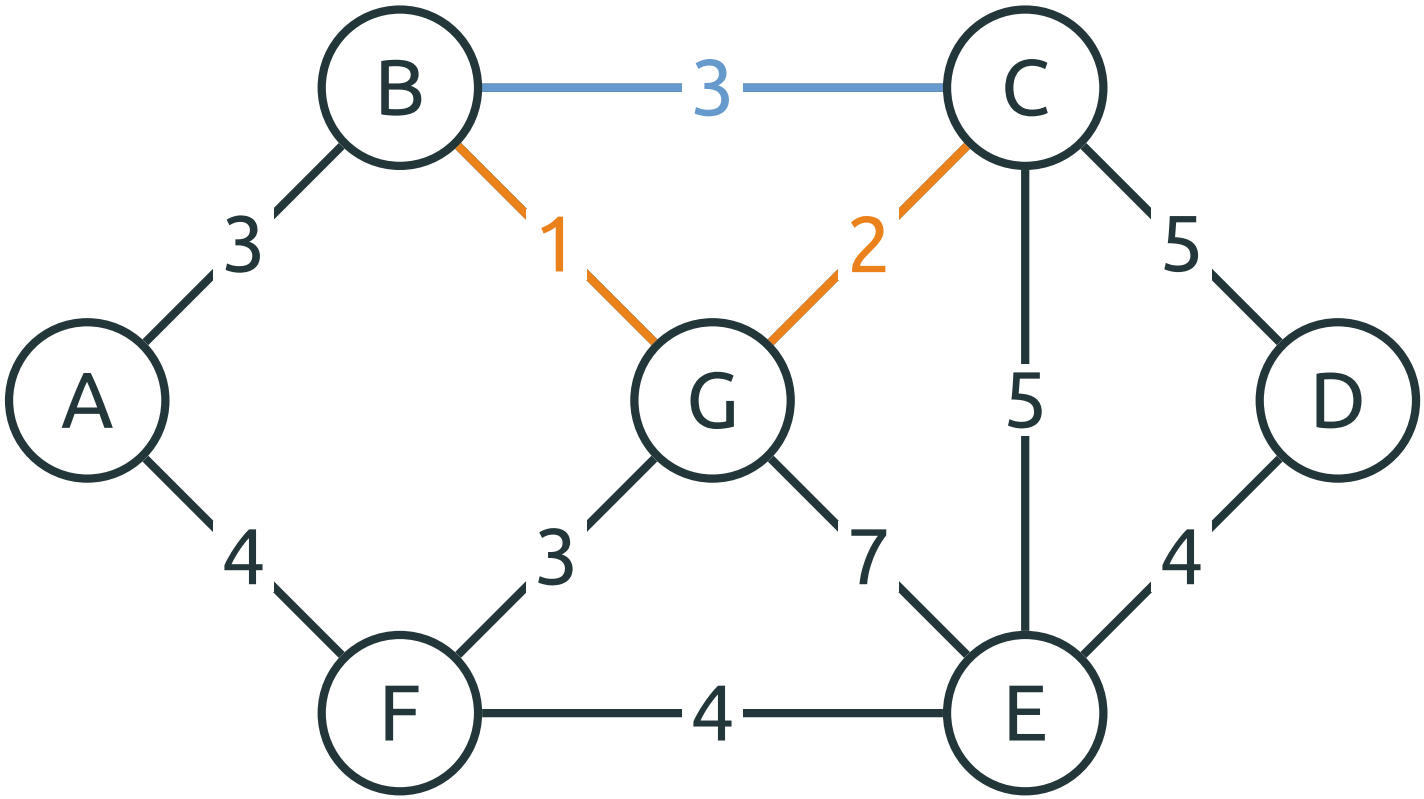
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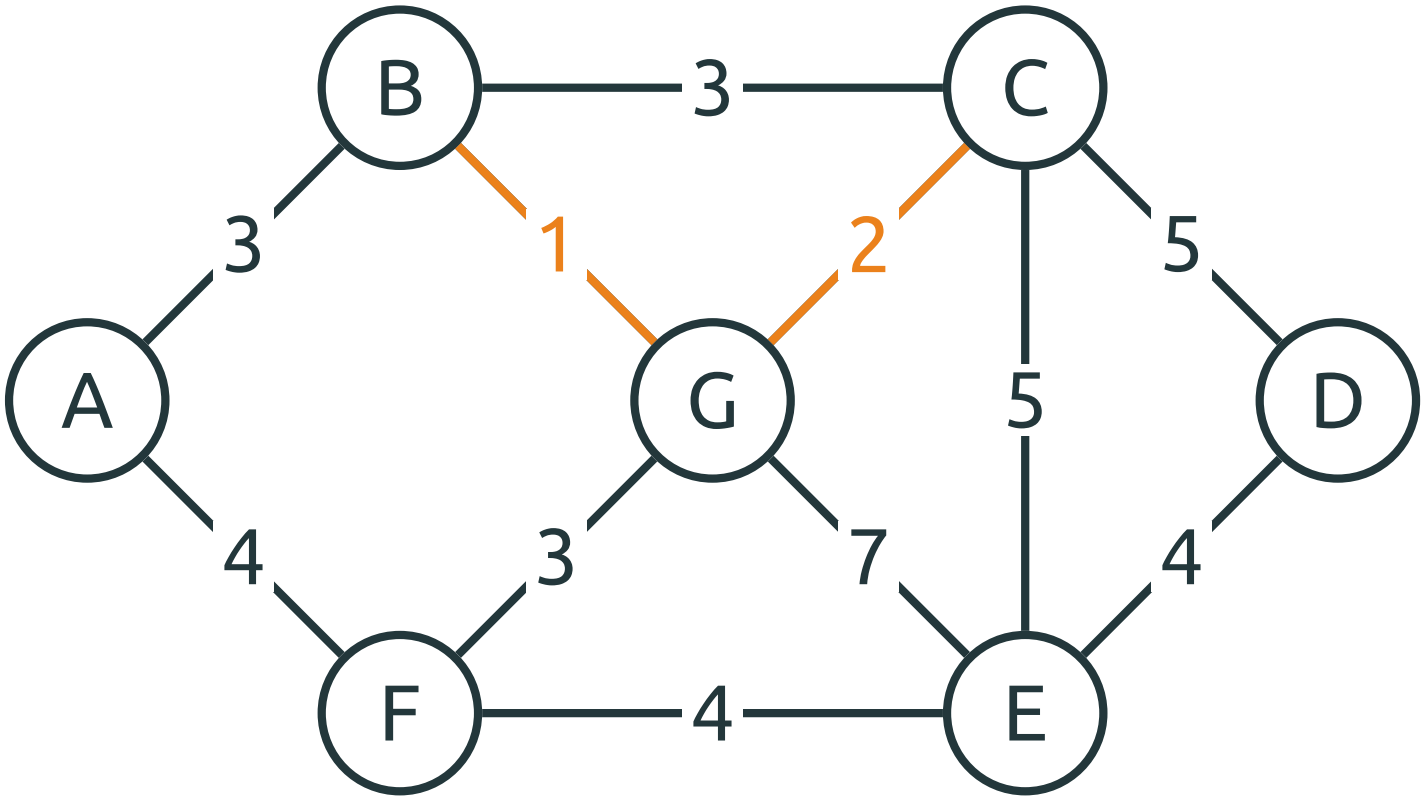
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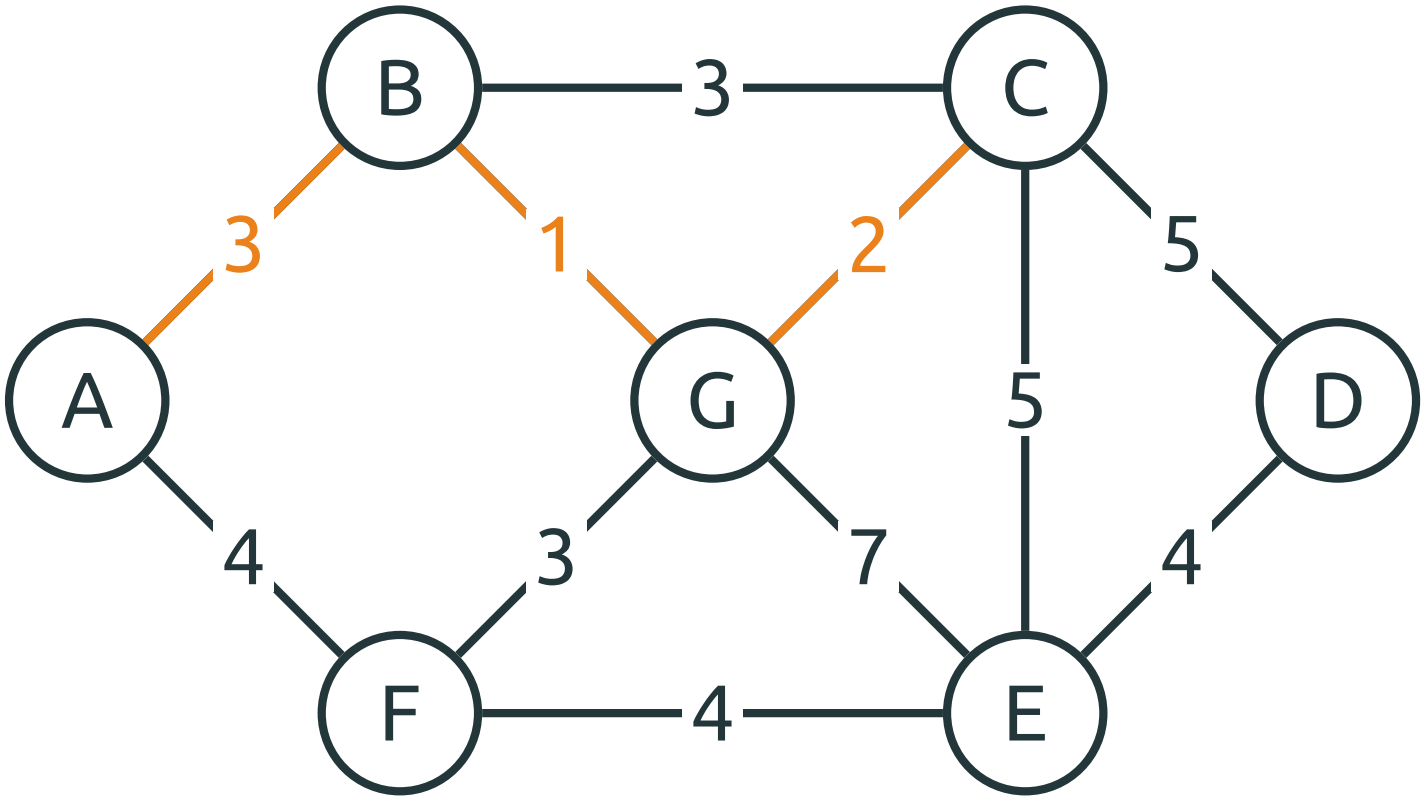
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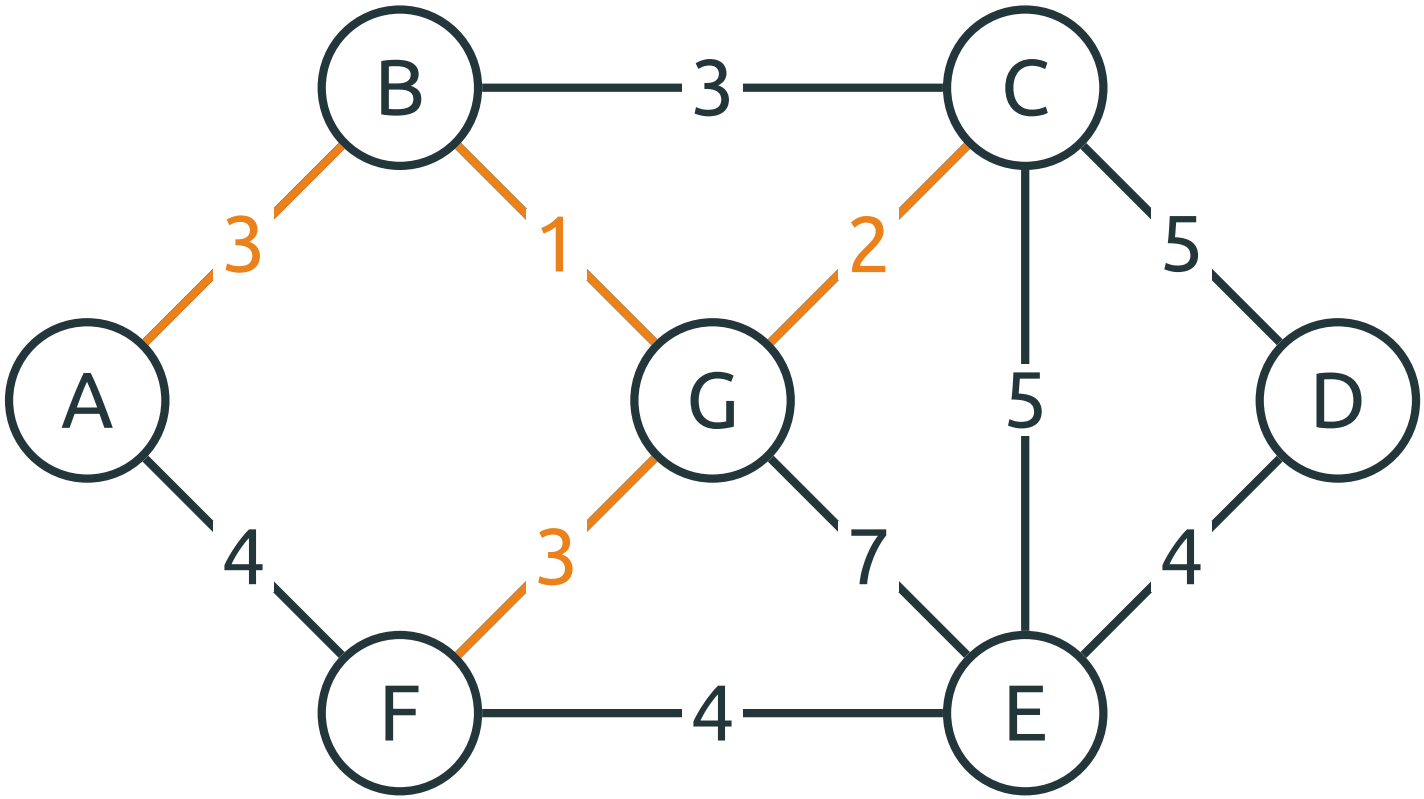
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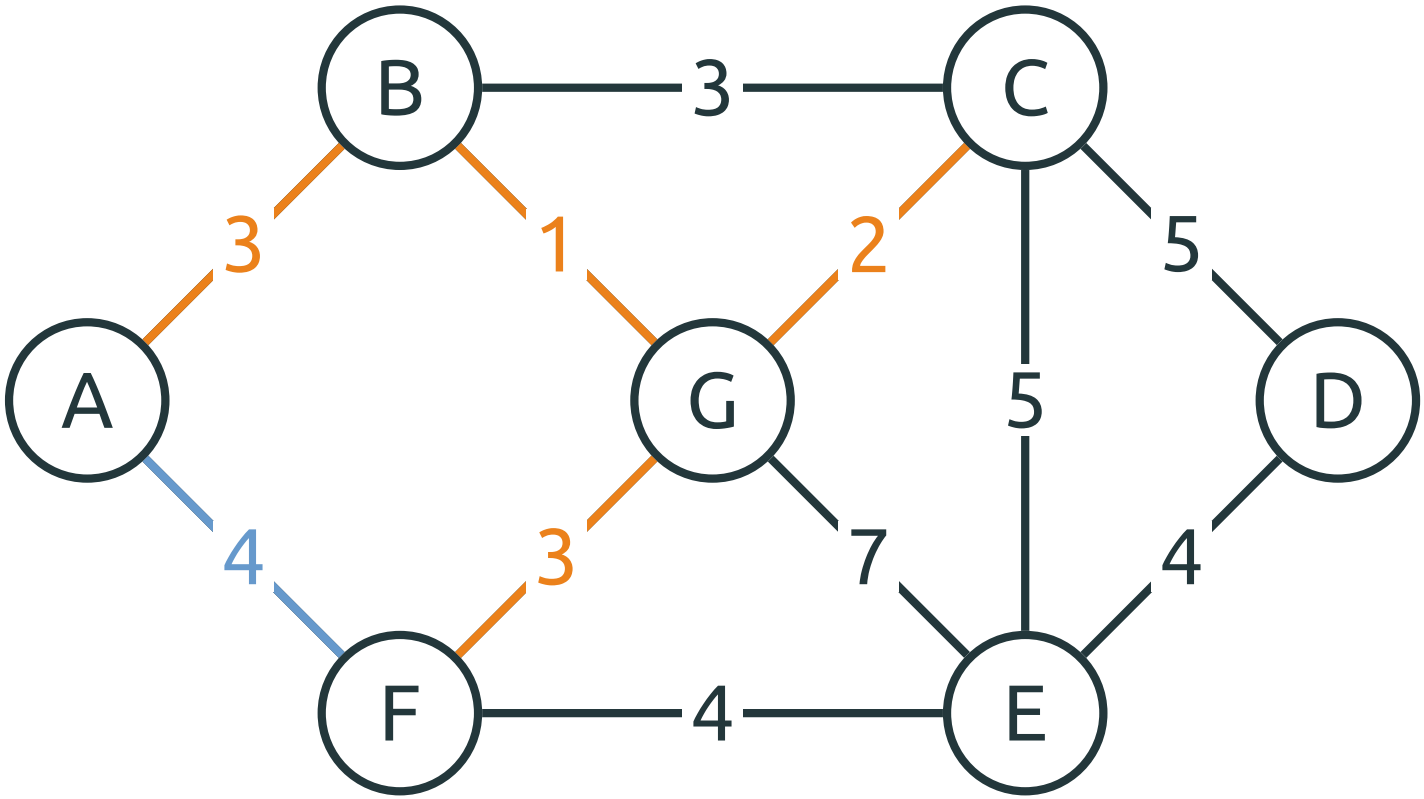
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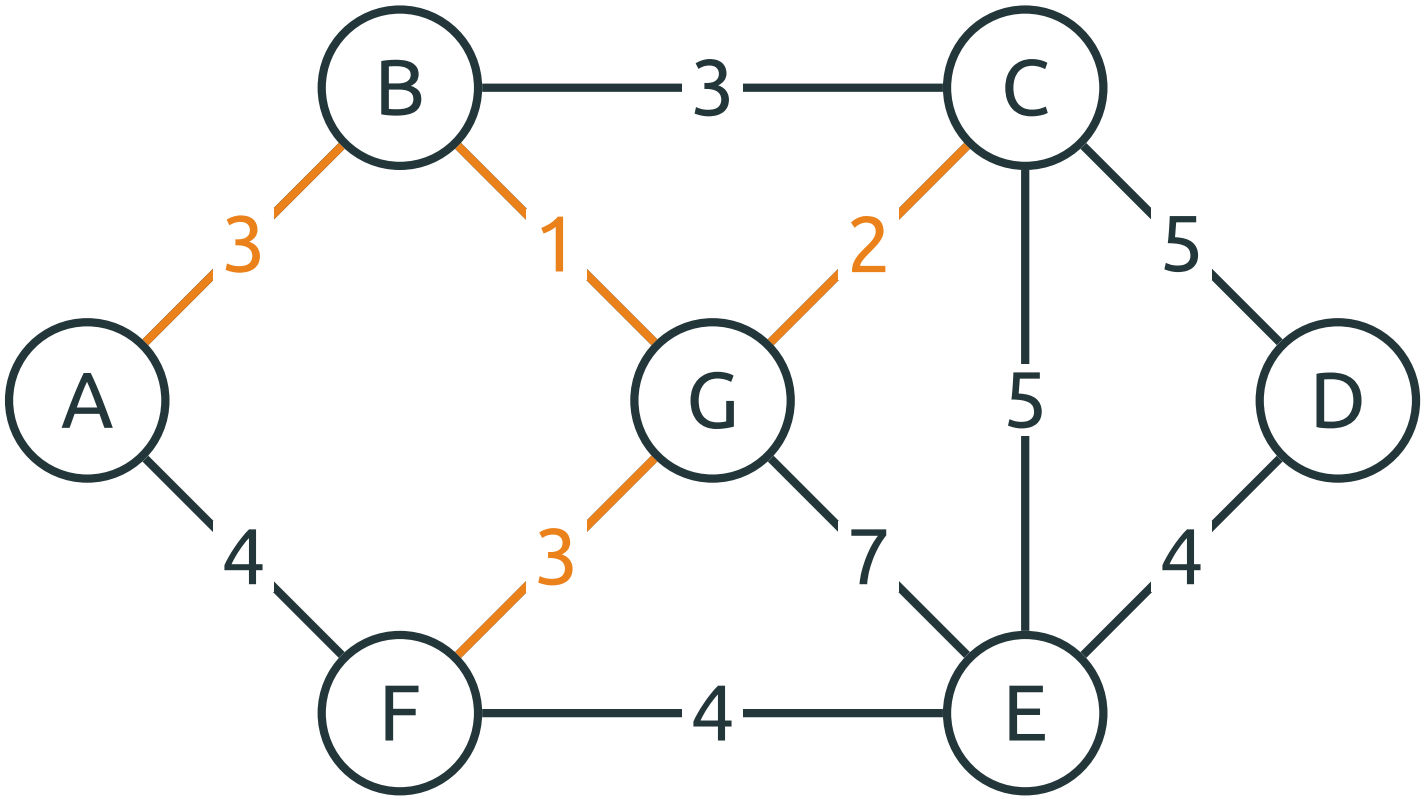
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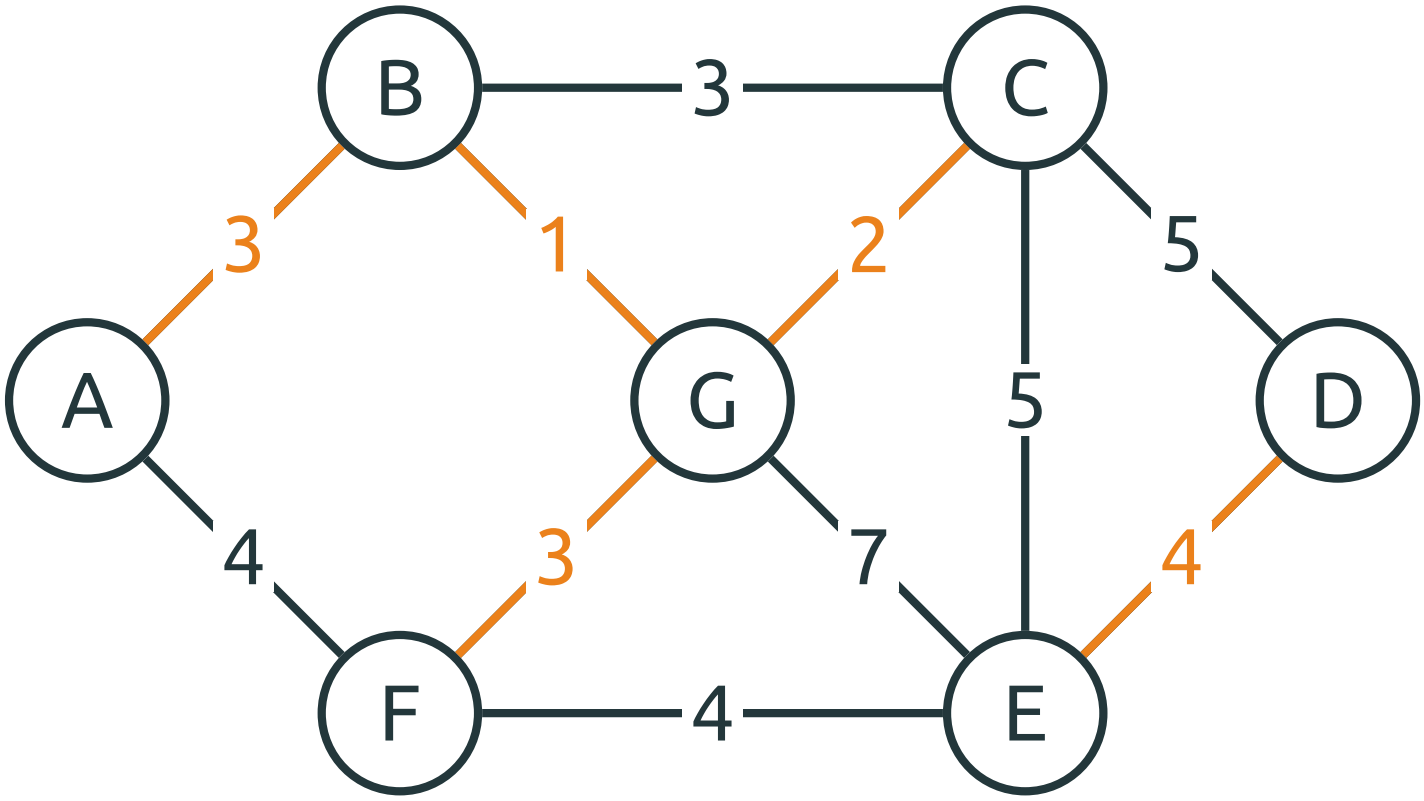
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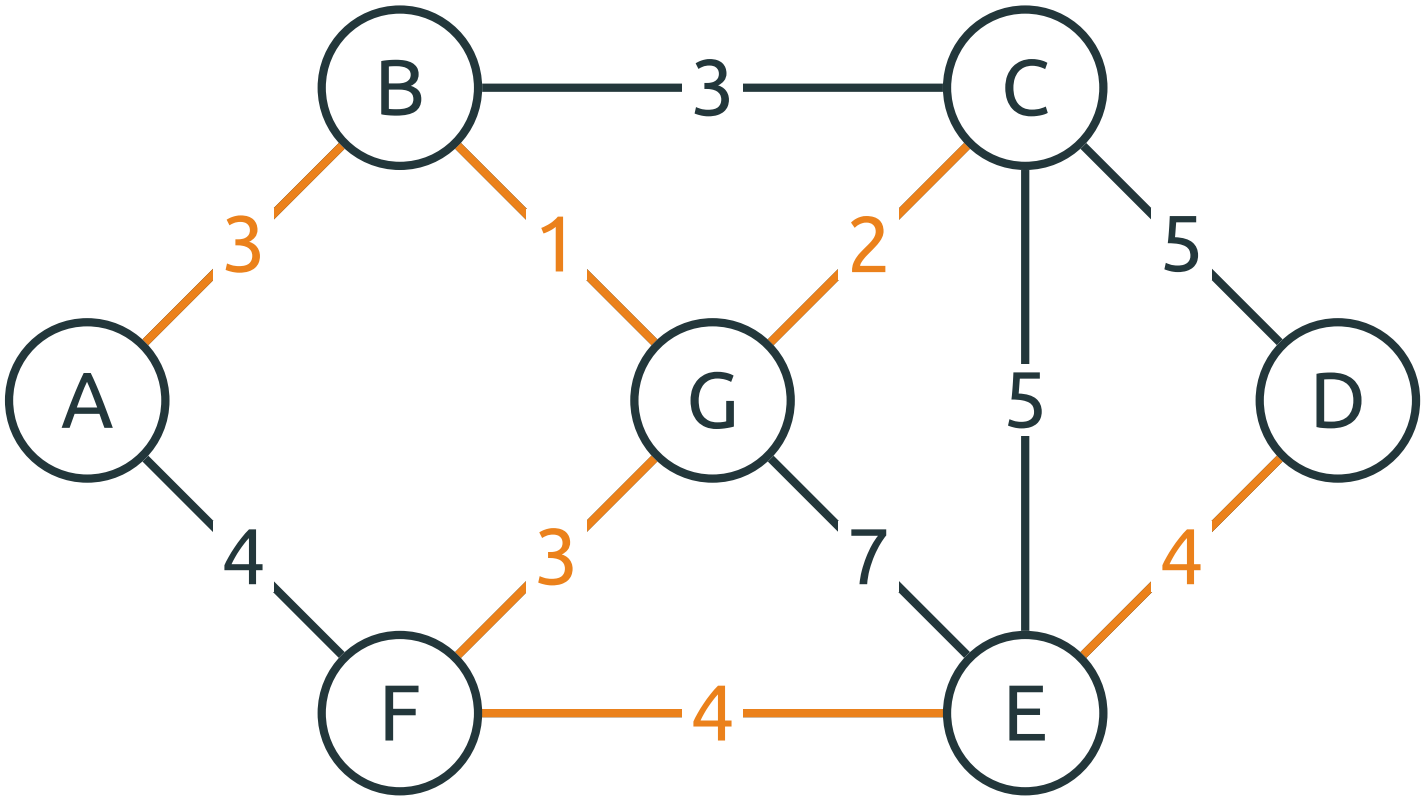
Road Repair



Road Repair



Road Repair



Outline

Road Repair

Trees

Minimum Spanning Tree

Definition

- A tree is a connected graph without cycles

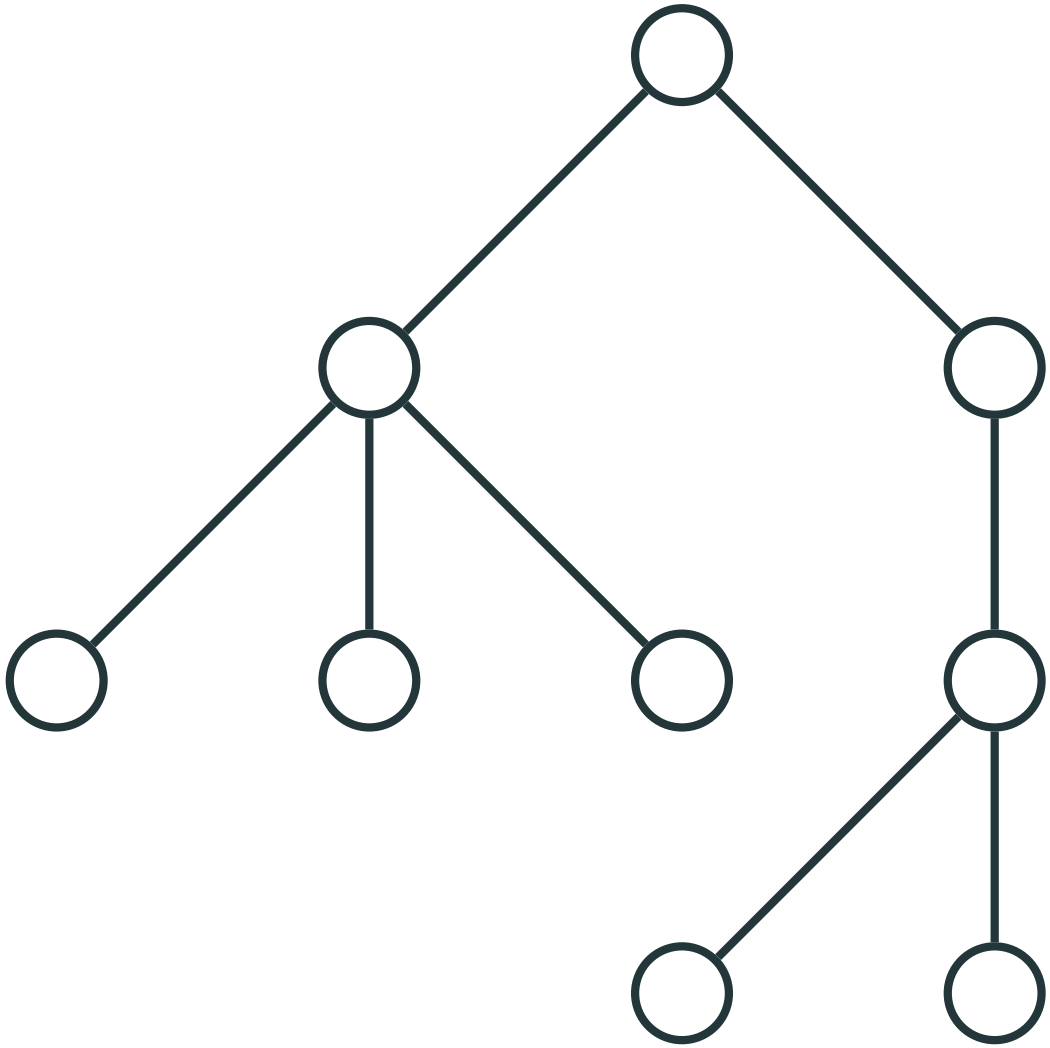
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Trees: Examples



Equivalent Definitions

- (I) A tree is a connected graph without cycles
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- We'll prove that (I) $\rightarrow$ (II) $\rightarrow$ (III) $\rightarrow$ (I)

(I) $\rightarrow$ (II)

- A connected graph on n vertices without cycles has $n - 1$ edges

(I) $\rightarrow$ (II)

- A connected graph on n vertices without cycles has $n - 1$ edges
- Induction on n

(I) $\rightarrow$ (II)

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- Base case. $n = 1$, 0 edges

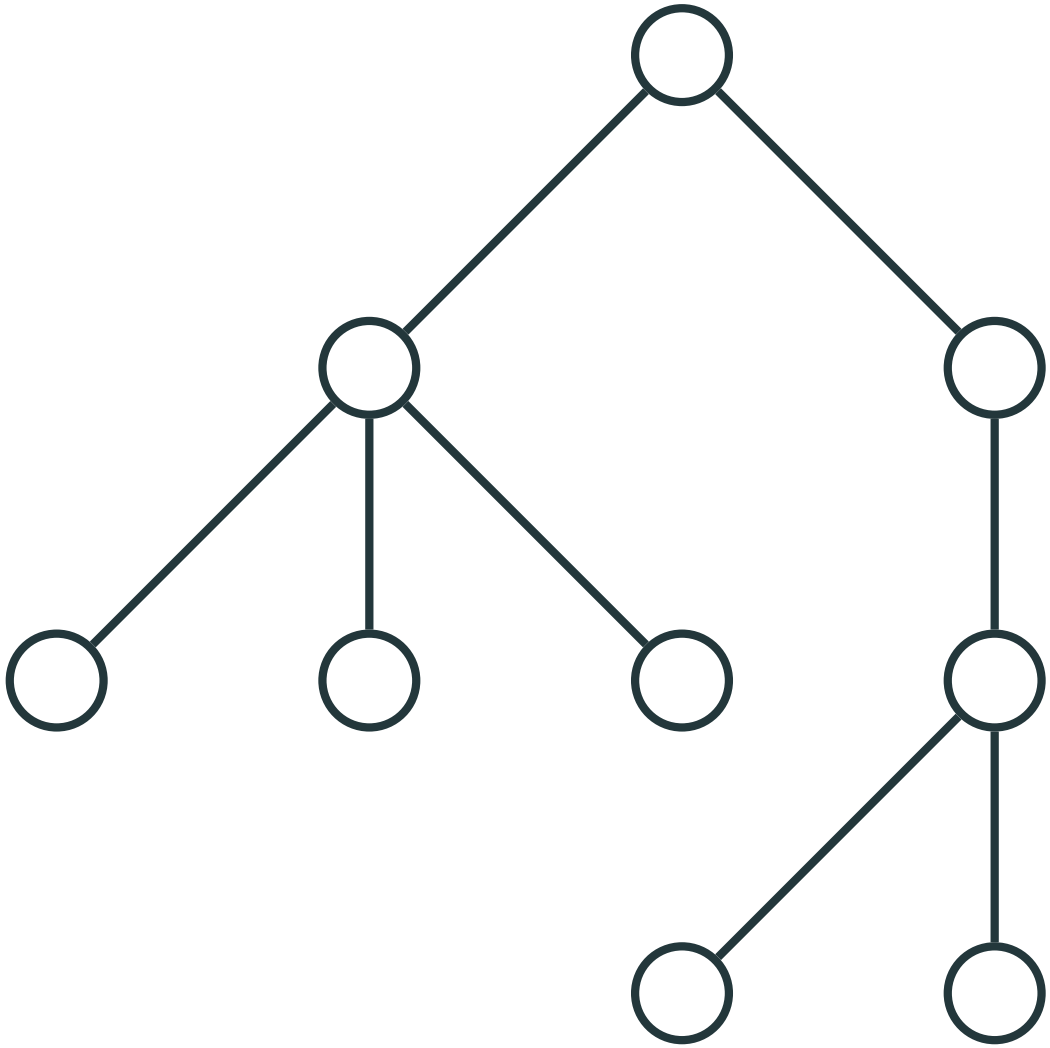
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- A connected graph on n vertices without cycles has $n - 1$ edges
- Induction on n
- Base case. $n = 1$, 0 edges
- Induction hypothesis. Every connected graph on $t \leq k$ vertices has $t - 1$ edges

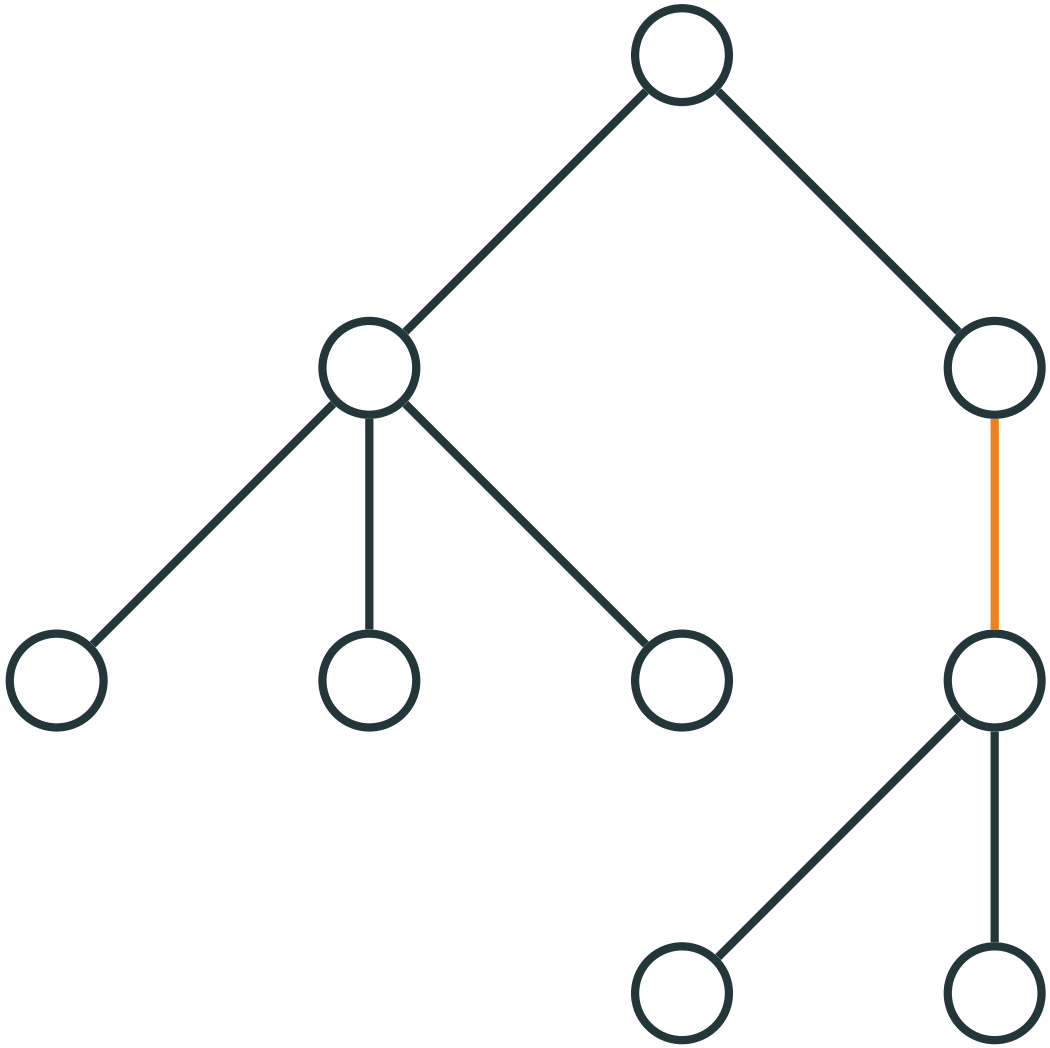
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- Induction step. Every connected graph on $k + 1$ vertices has k edges

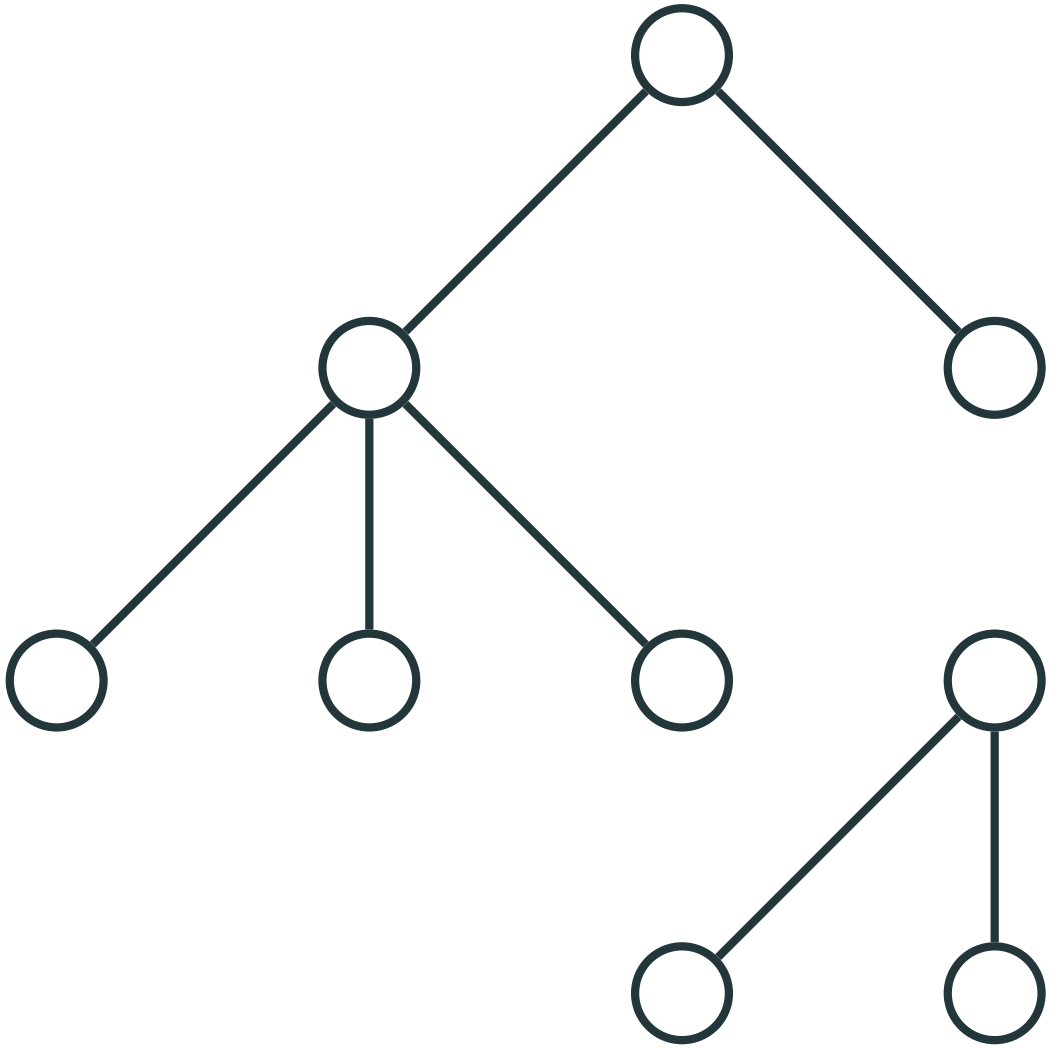
(I) $\rightarrow$ (II)



(I) → (II)



(I) $\rightarrow$ (II)



(I) $\rightarrow$ (II)

- Remove an edge

(I) $\rightarrow$ (II)

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- Thus, the original graph has $(n_1 - 1) + (n_2 - 1) + 1 = n - 1$ edges

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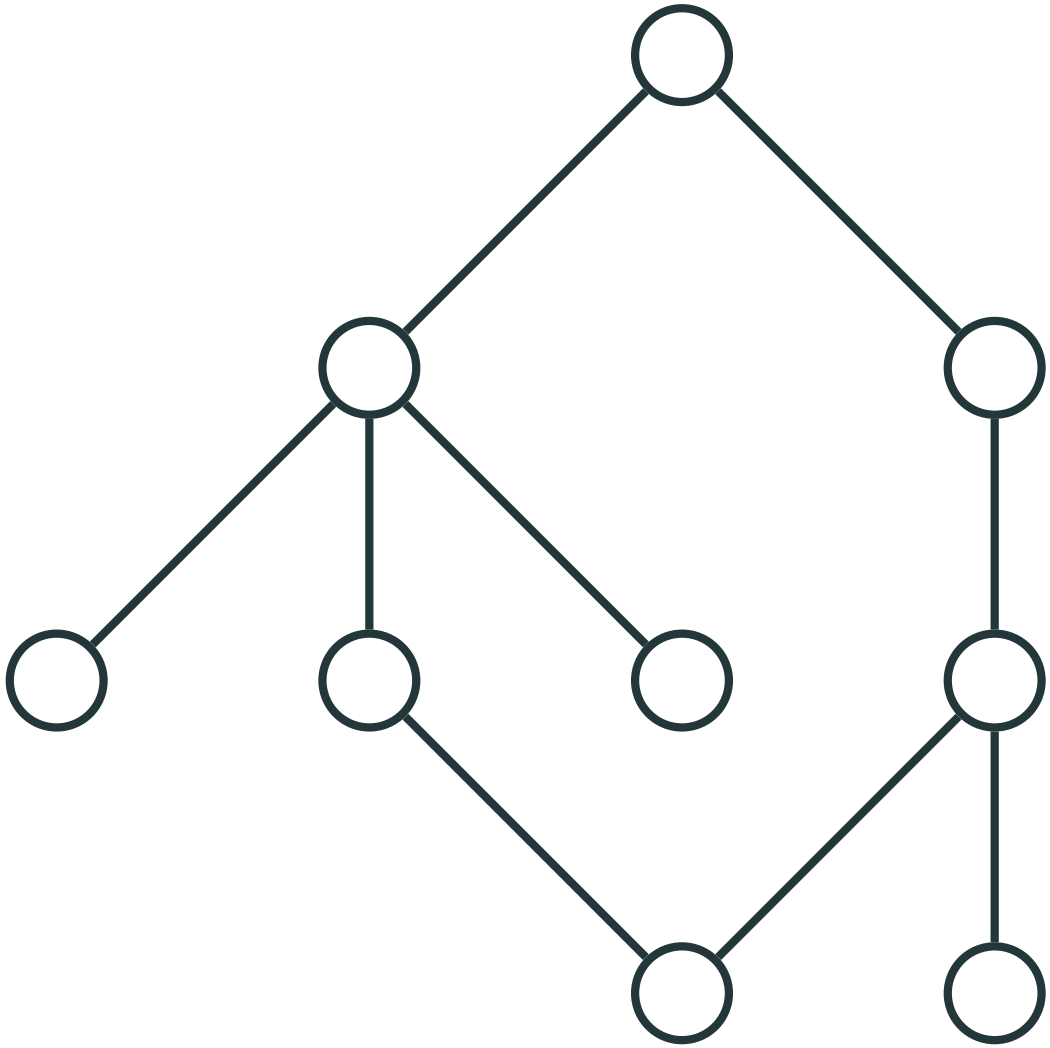
(II) $\rightarrow$ (III)

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- Then the number of edges is n

(II) $\rightarrow$ (III)



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Outline

Road Repair

Trees

Minimum Spanning Tree

Spanning Trees

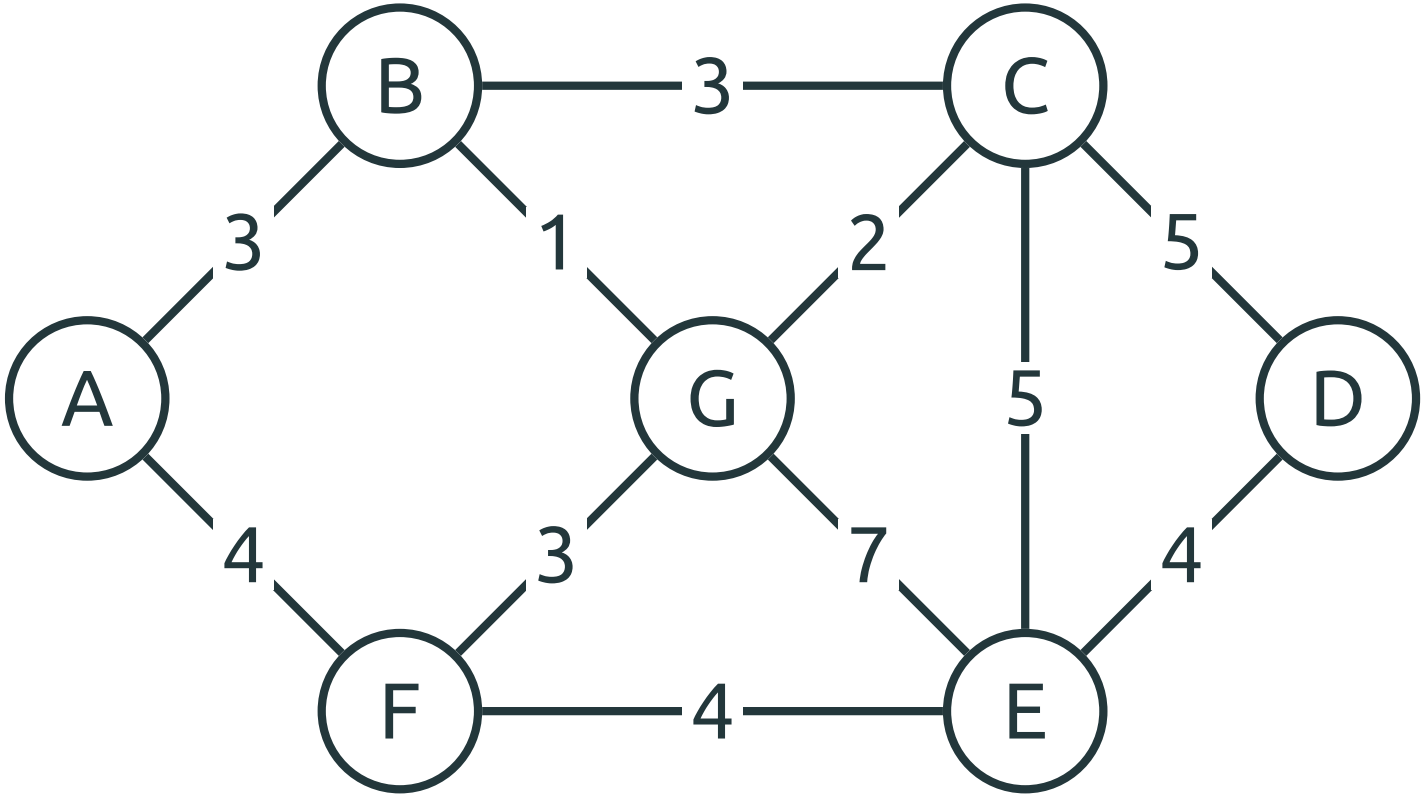
- A Spanning Tree of a graph *G*, is a subgraph of *G* which is a tree and contains all vertices of *G*

Spanning Trees

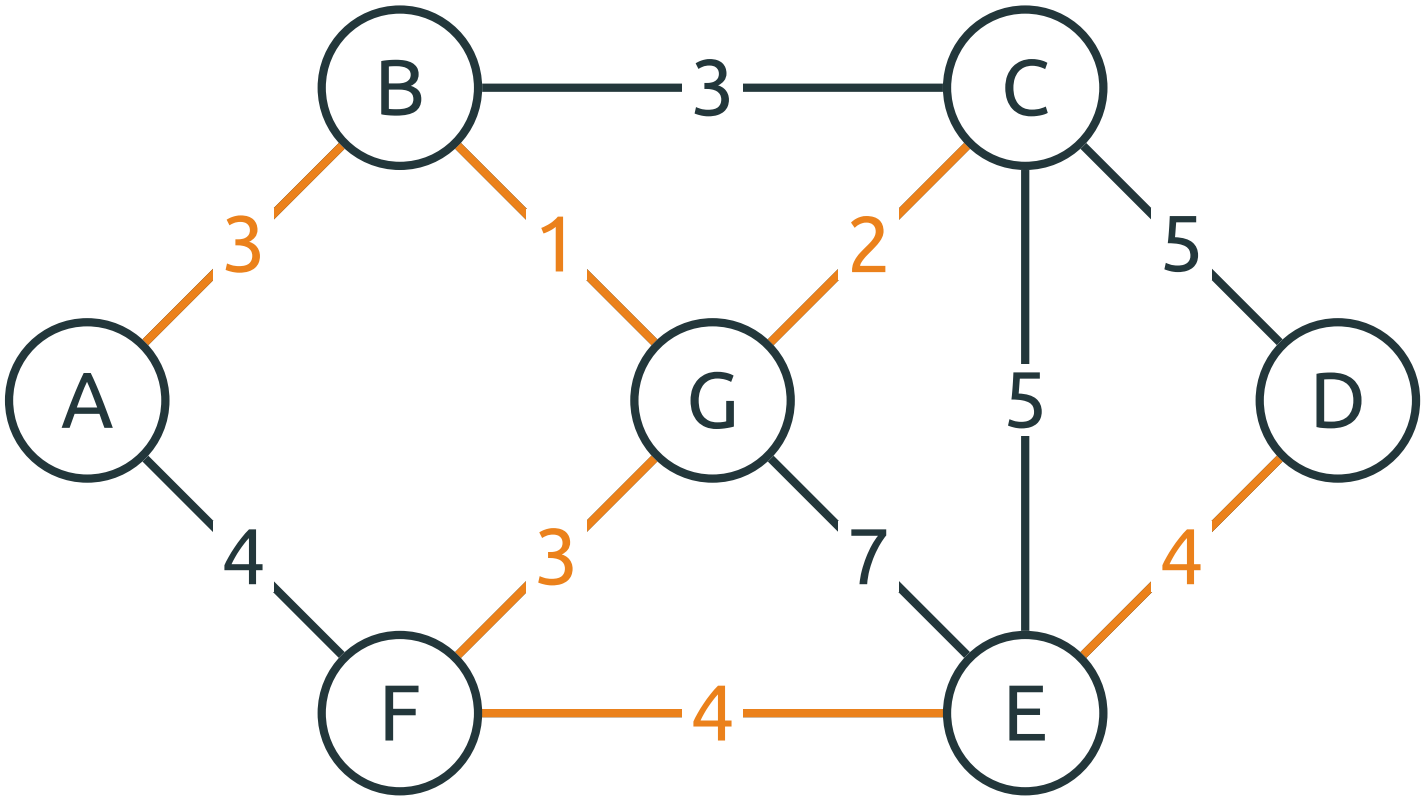
- A Spanning Tree of a graph G, is a subgraph of G which is a tree and contains all vertices of G

- A Minimum Spanning Tree of a weighted graph G is a spanning tree of the smallest weight

Minimum Spanning Tree: Examples



Minimum Spanning Tree: Examples



Kruskal's Algorithm

- Start with an empty graph T

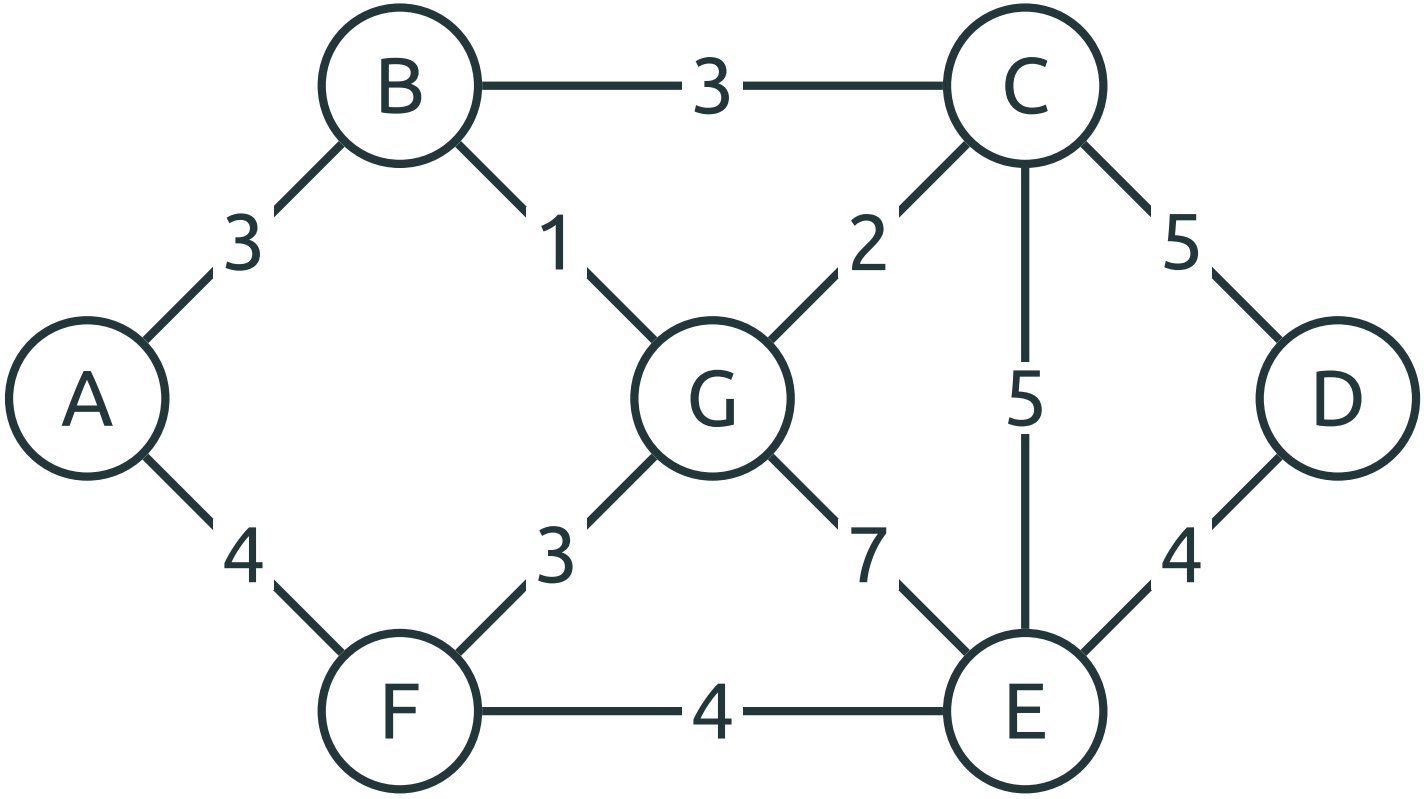
Kruskal's Algorithm

- Start with an empty graph T
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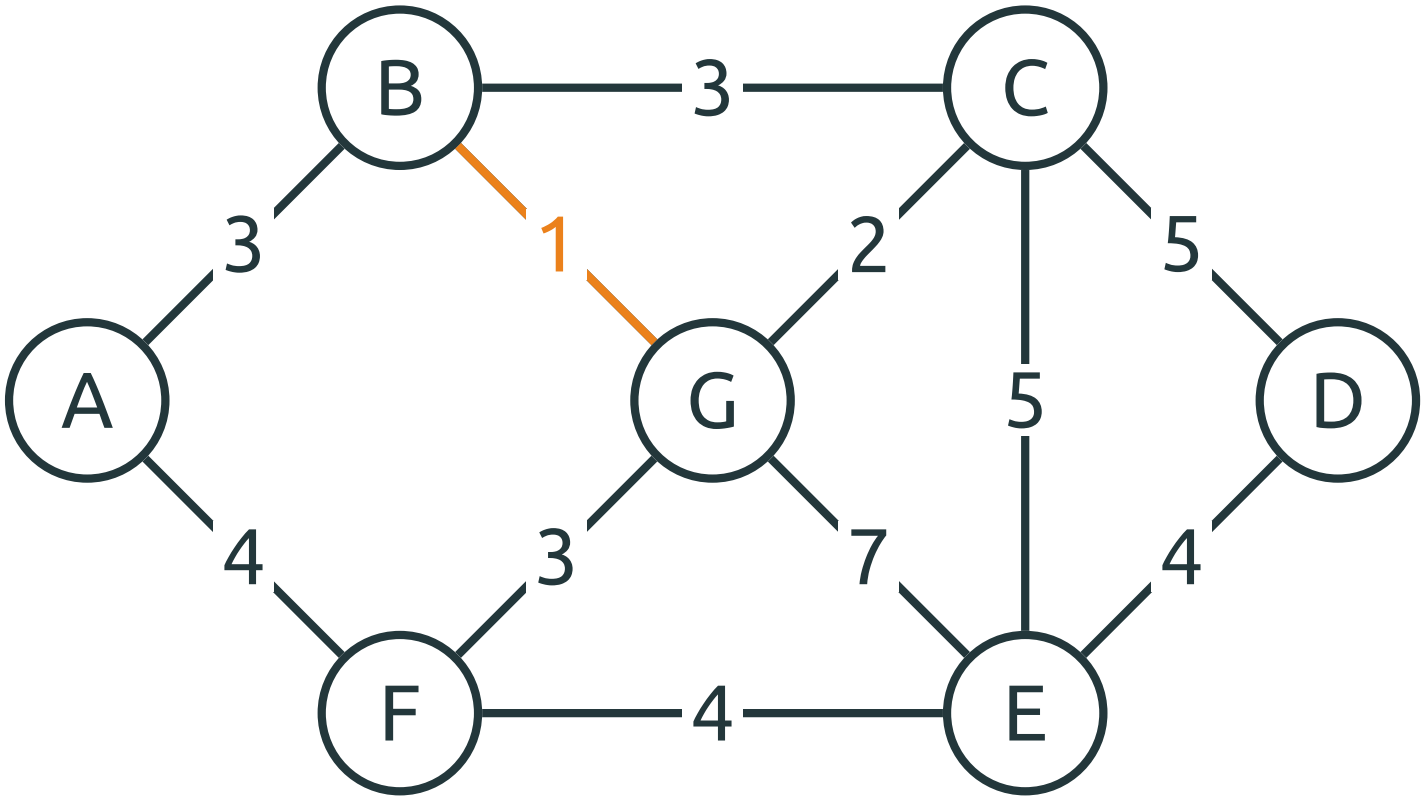
Kruskal's Algorithm

- Start with an empty graph T
- Repeat $n - 1$ times:
- Add to T an edge of the smallest weight which doesn't create a cycle in T

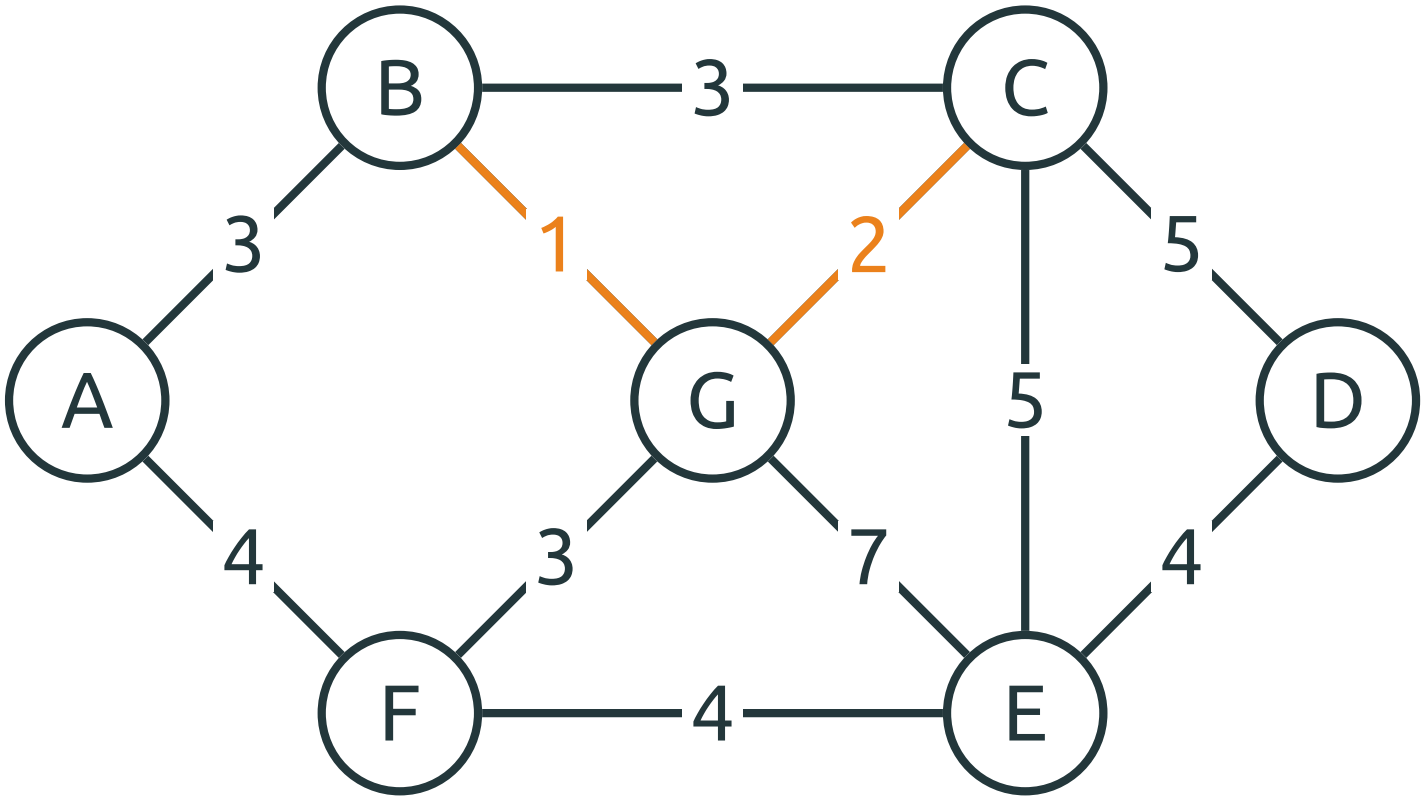
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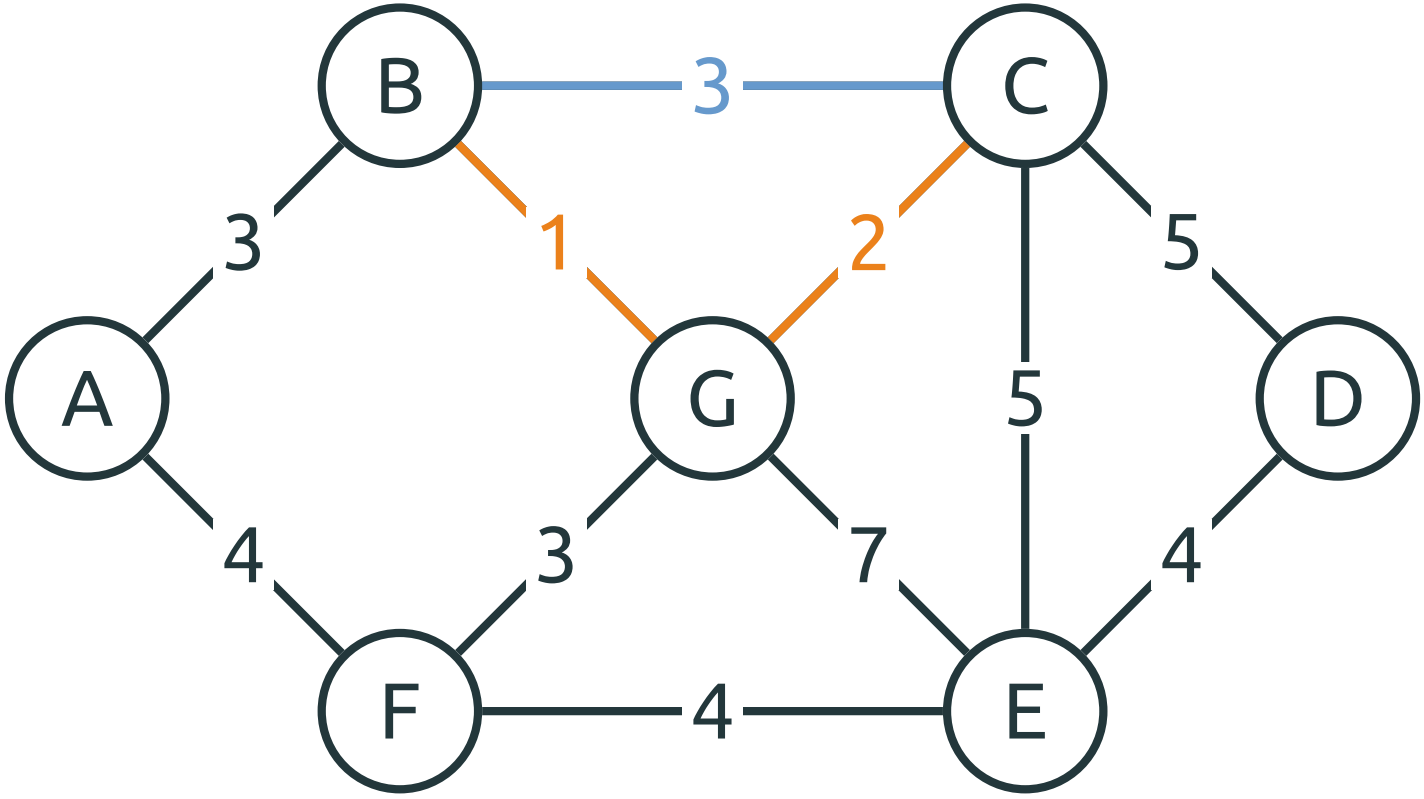
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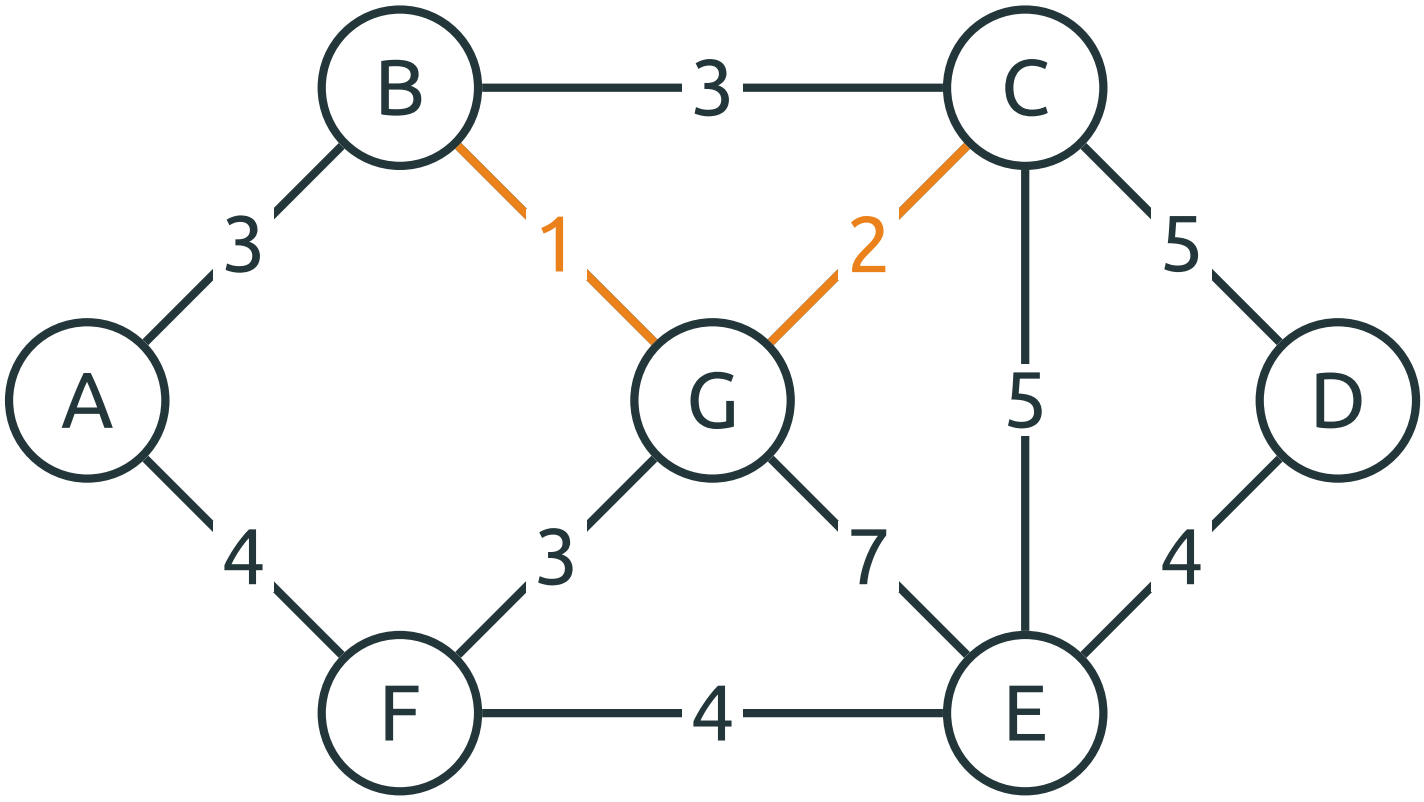
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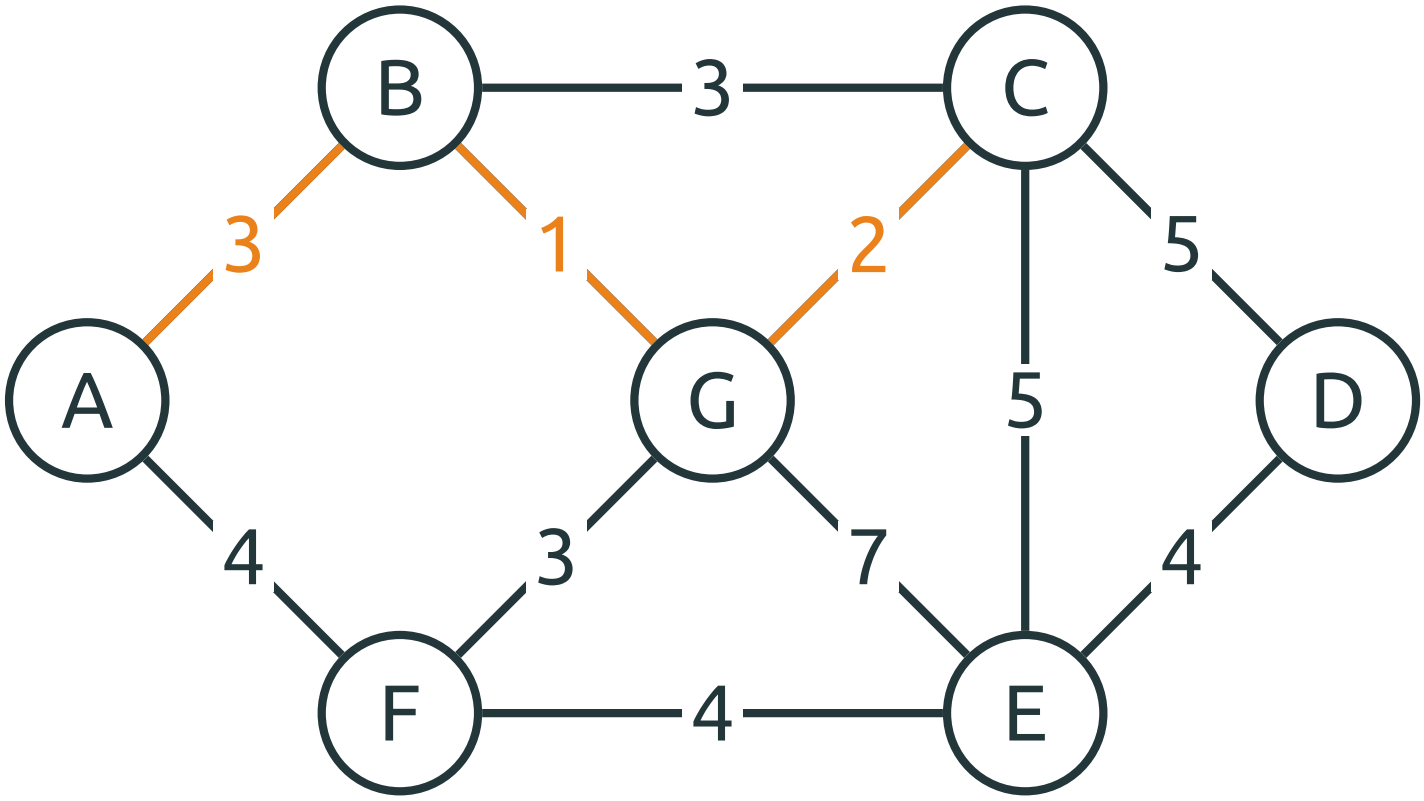
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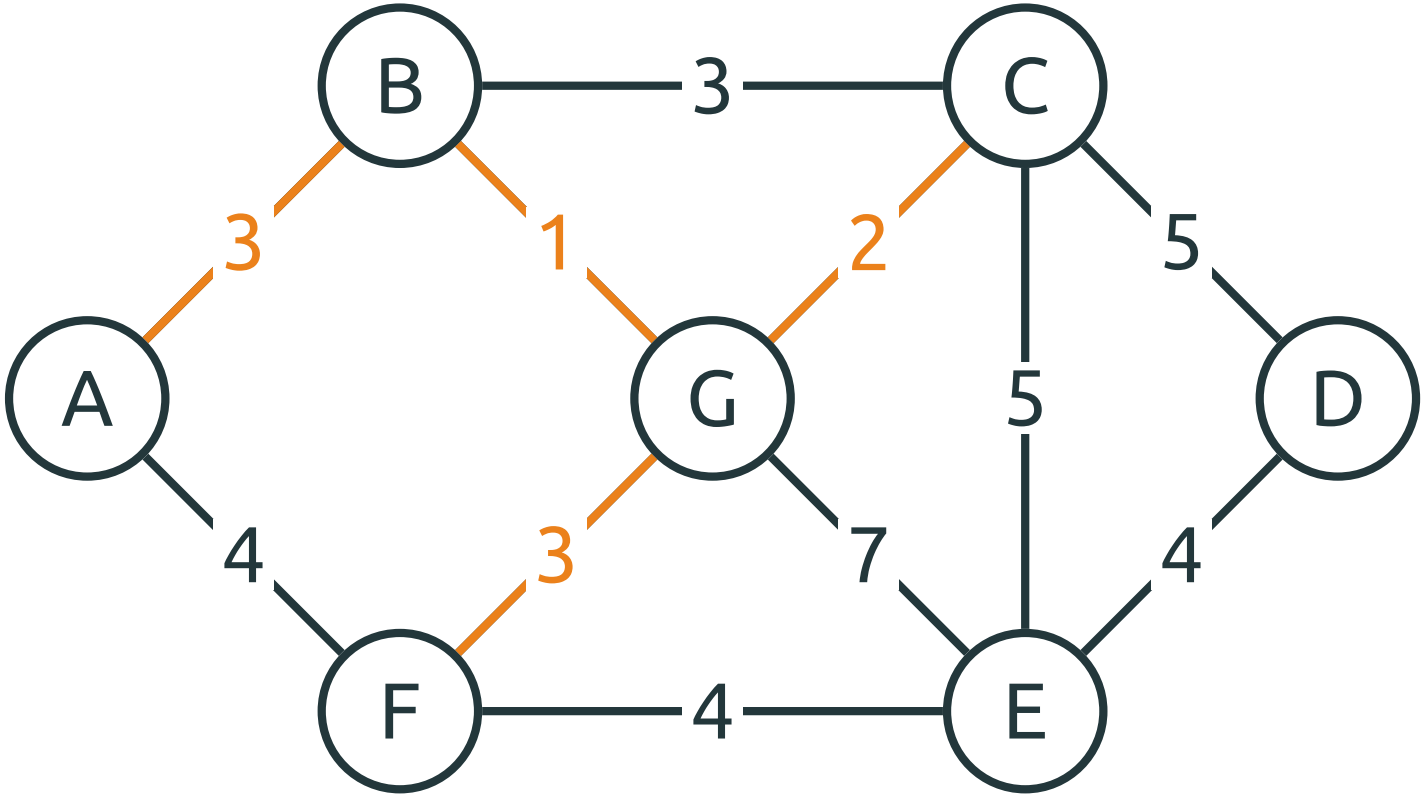
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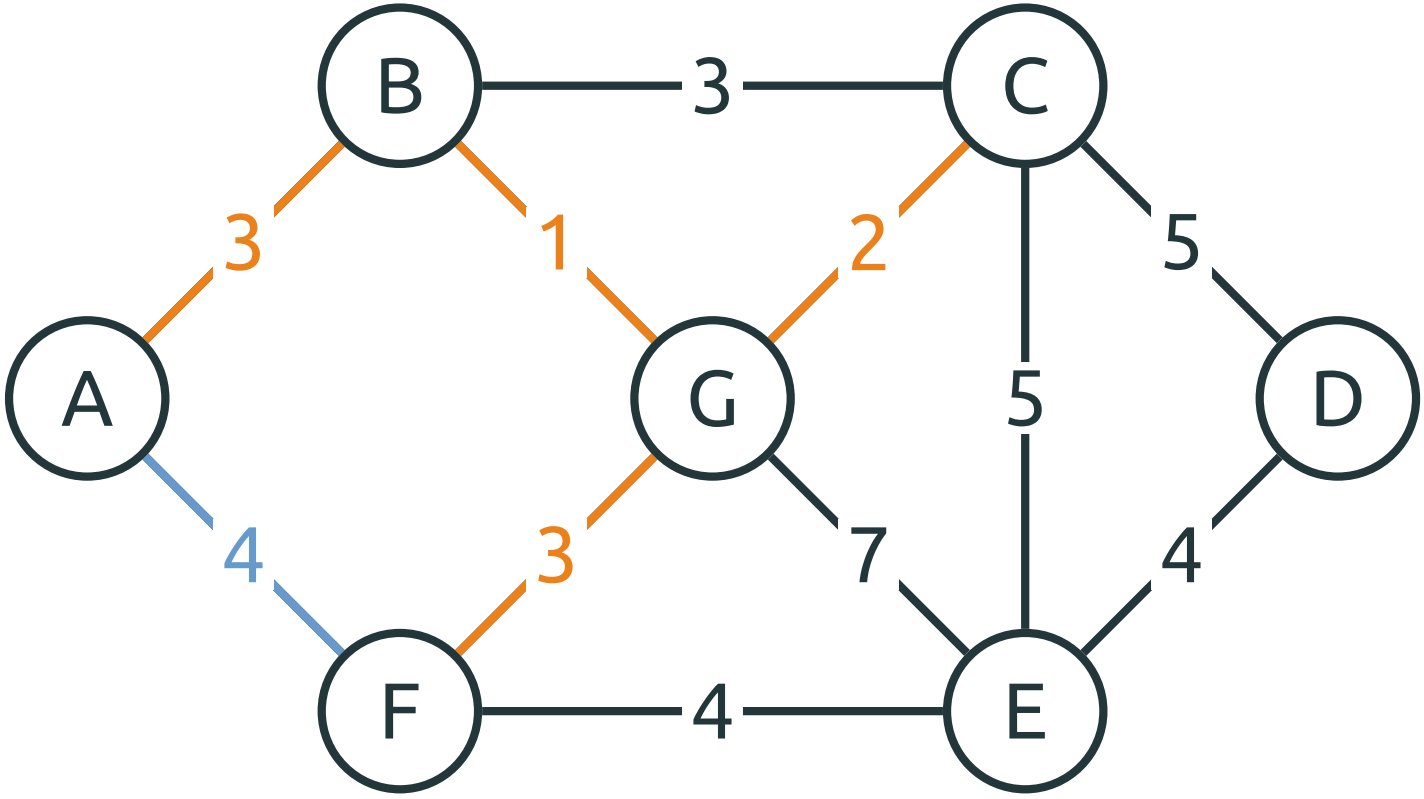
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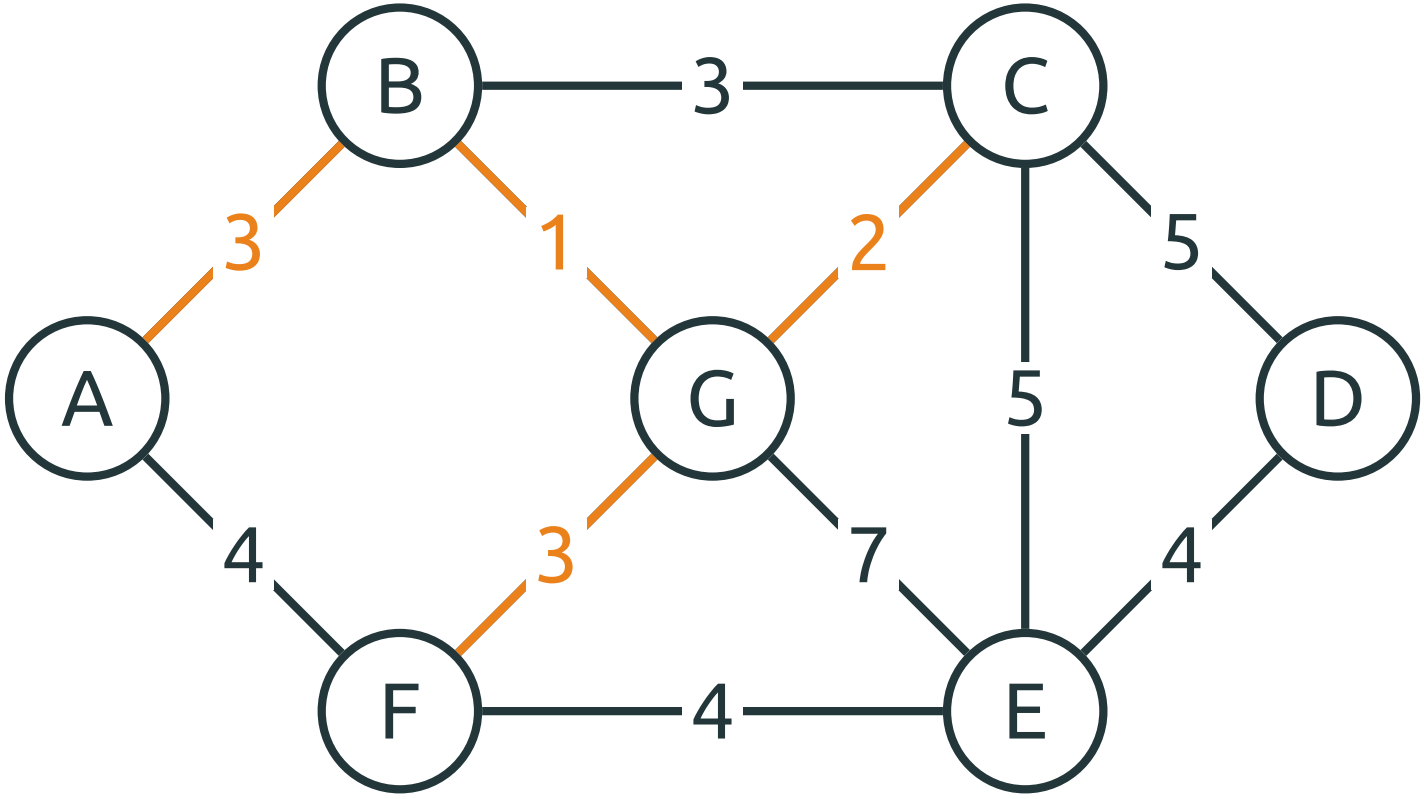
Kruskal's Algorithm: Examples



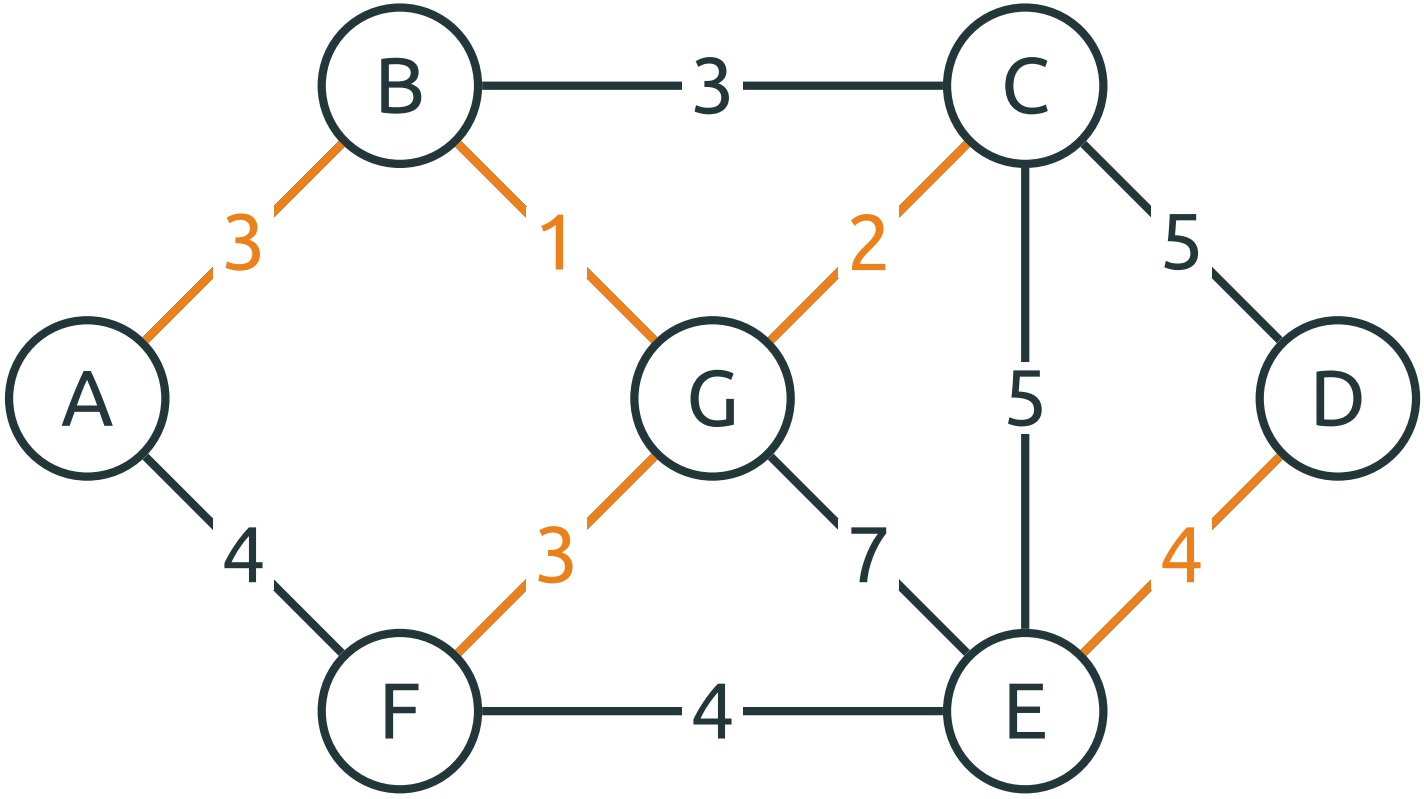
Kruskal's Algorithm: Examples



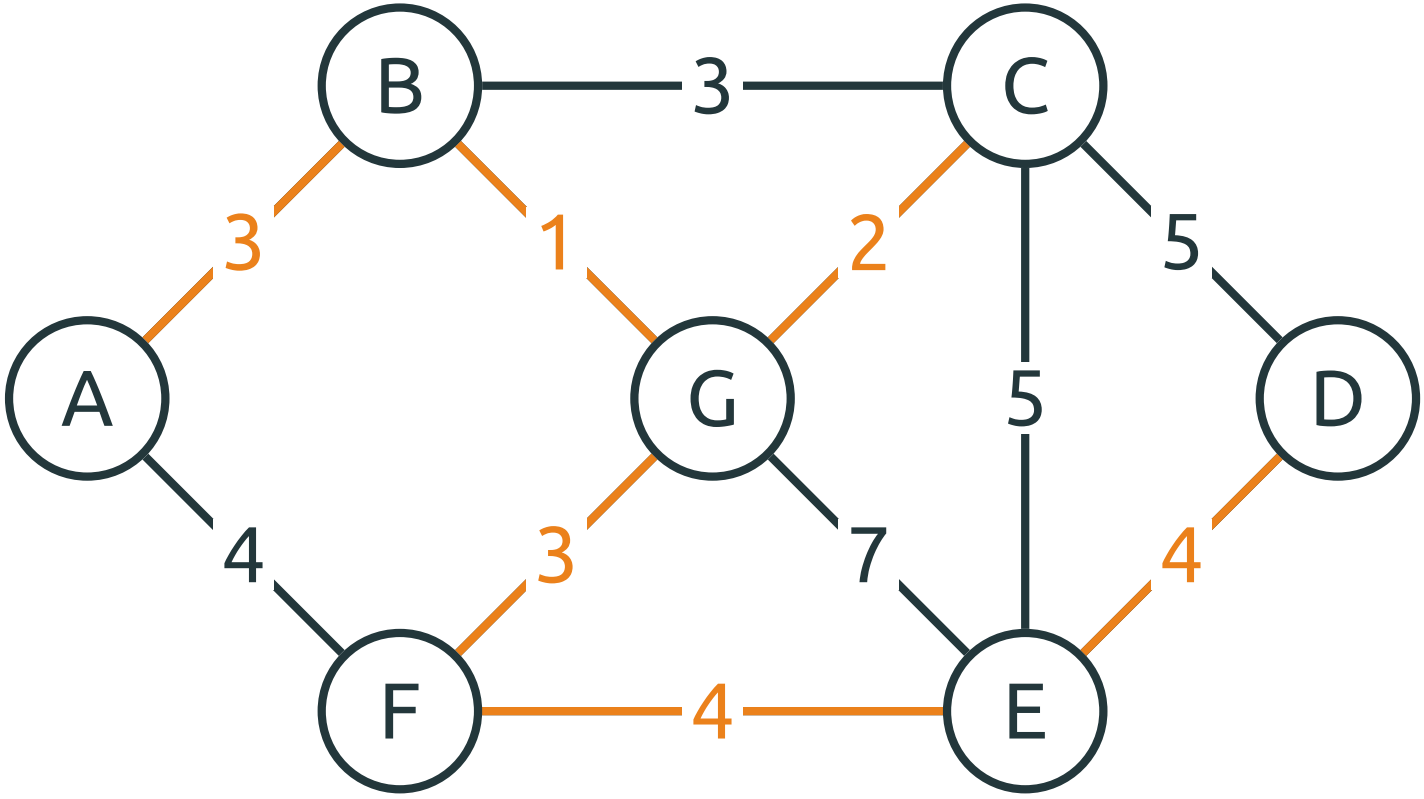
Kruskal's Algorithm: Examples



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Kruskal's Algorithm: Examples



Kruskal's Algorithm: Proof

- We'll show that at every step of the algorithm, there exists a Minimum Spanning Tree which contains all currently chosen edges

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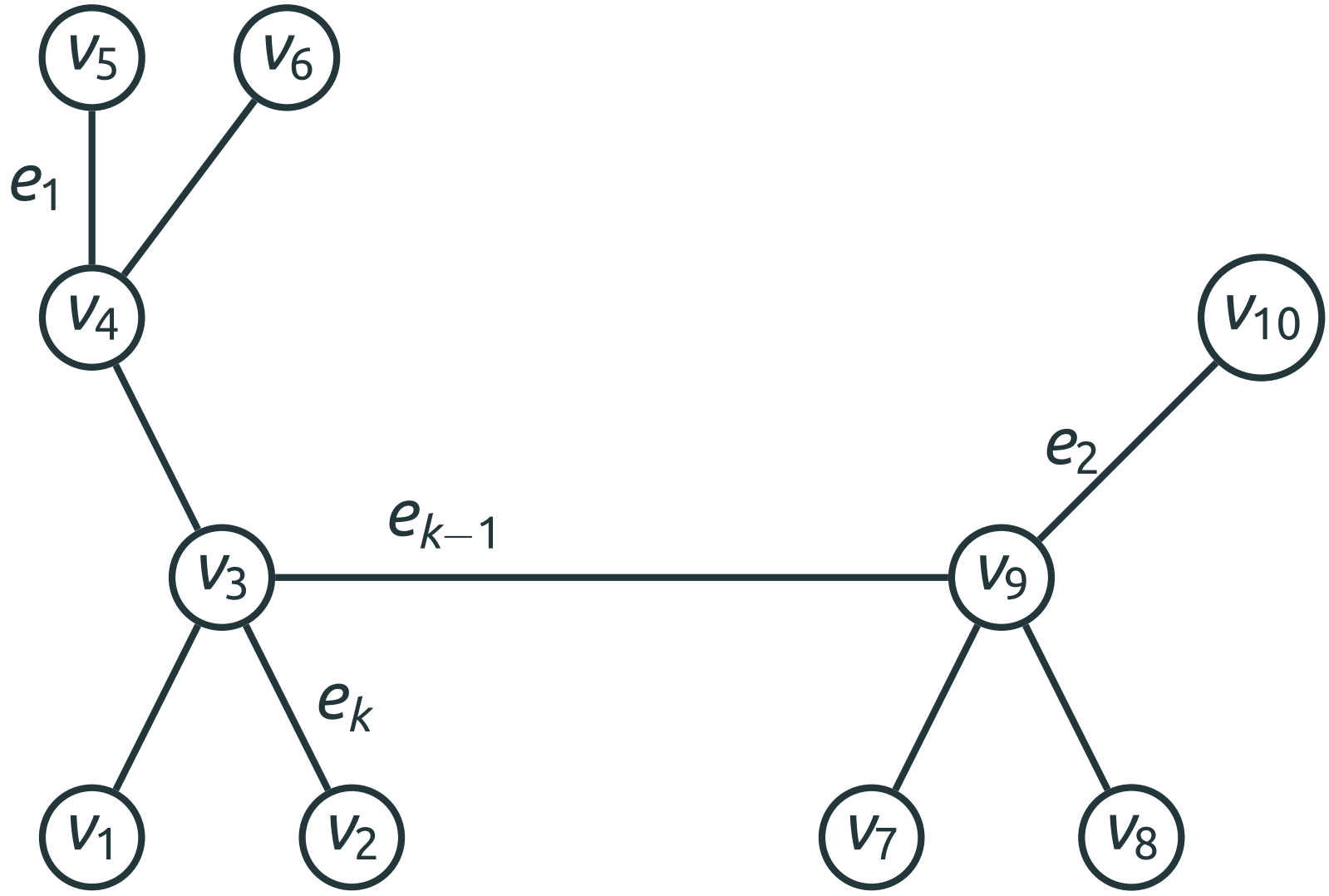
Kruskal's Algorithm: Proof

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- Base Case. $s = 0$, empty tree
- Induction hypothesis. True for step $s = k$

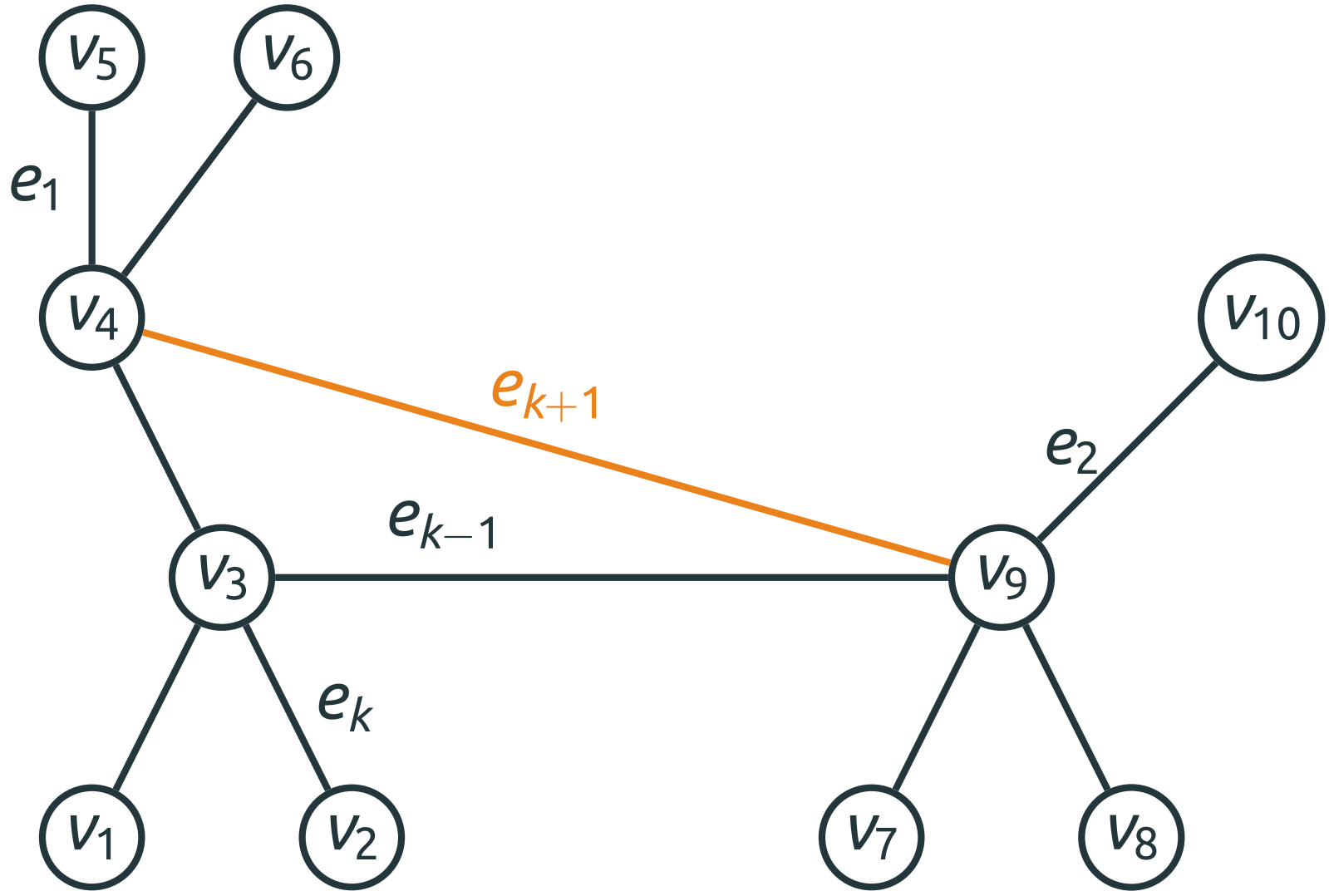
Kruskal's Algorithm: Proof

- We'll show that at every step of the algorithm, there exists a Minimum Spanning Tree which contains all currently chosen edges
- Induction on Step Number s
- Base Case. $s = 0$, empty tree
- Induction hypothesis. True for step $s = k$
- Induction step. We'll show that there exists a Minimum Spanning Tree which contains the first $k + 1$ edges of T

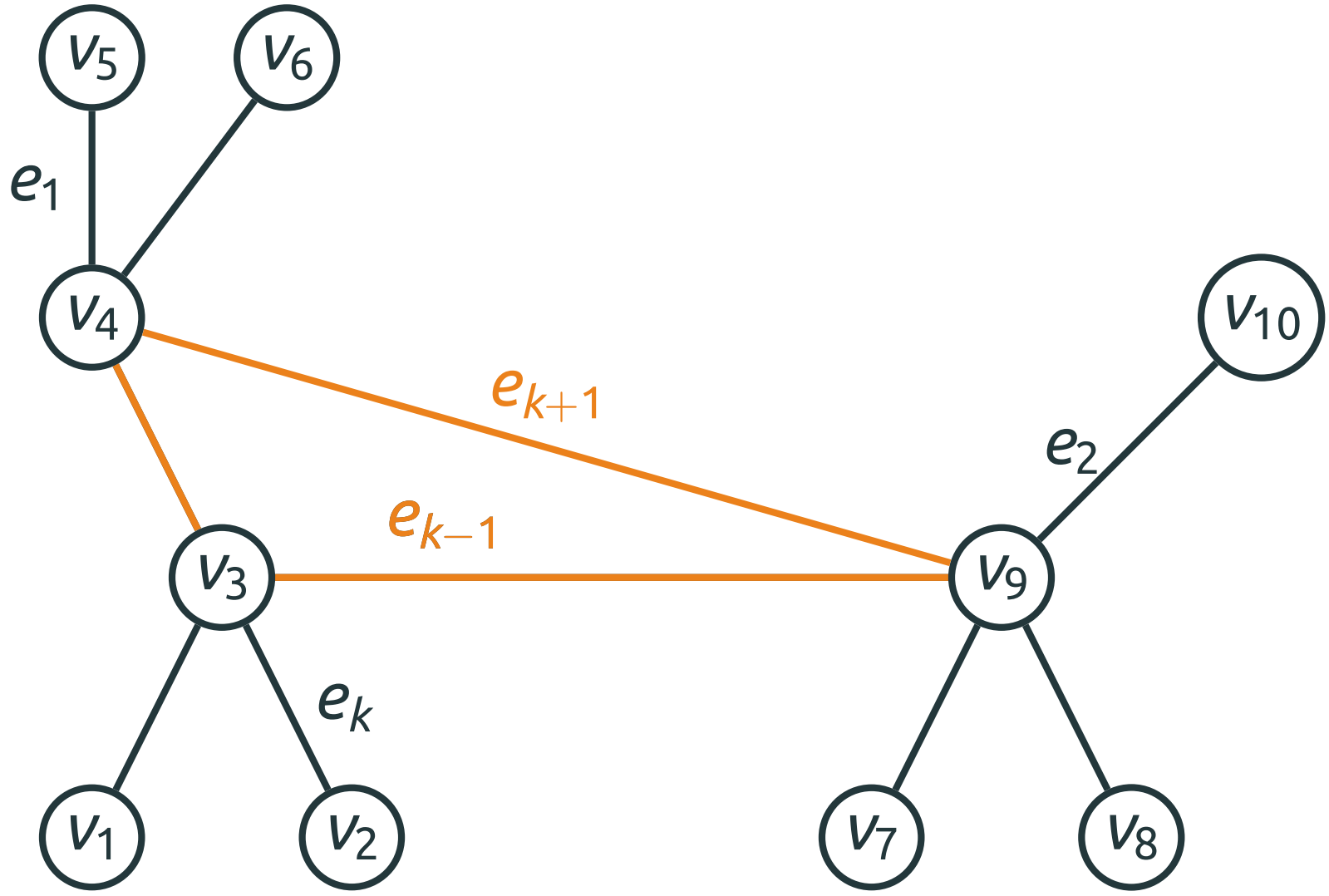
Kruskal's Algorithm: Proof



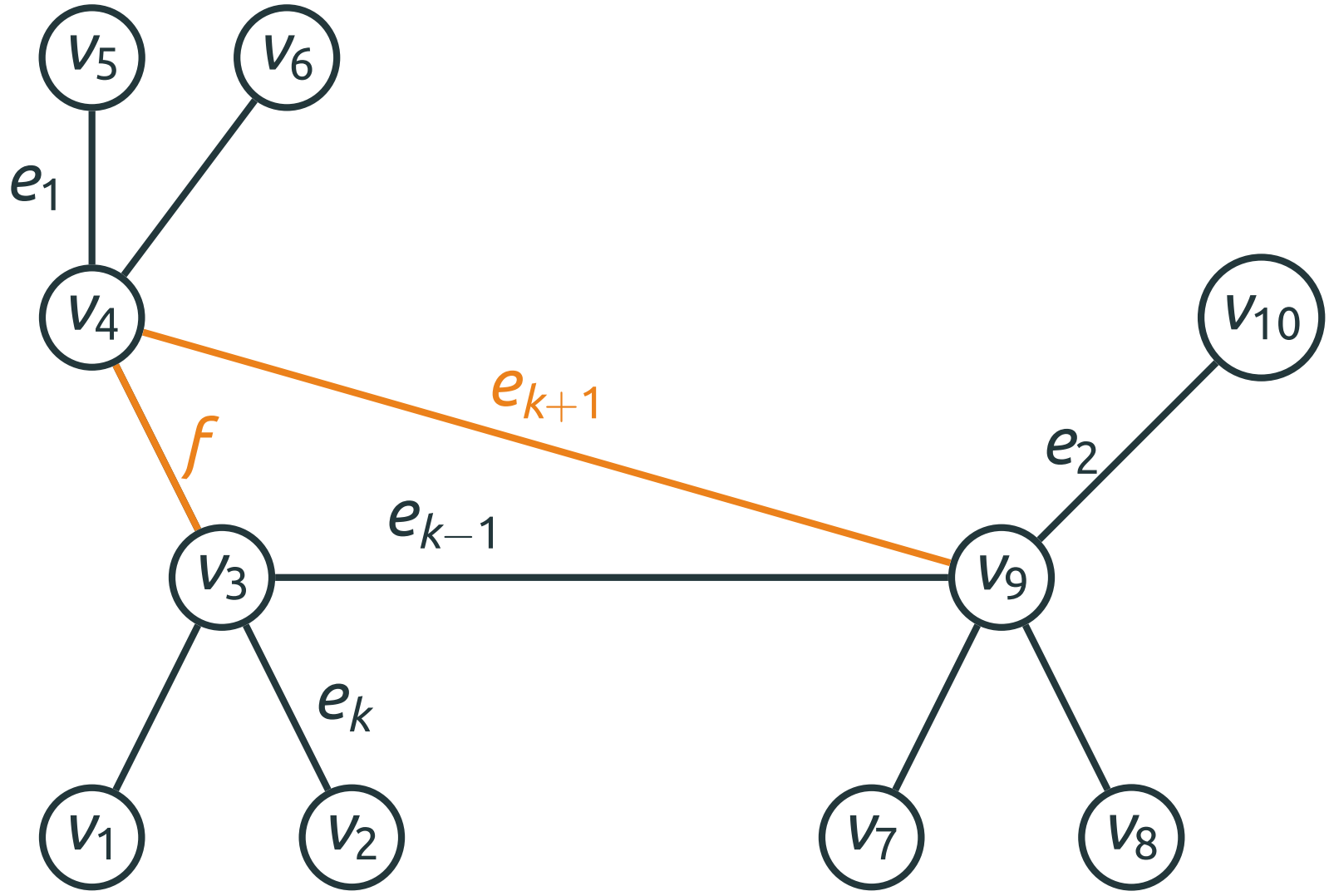
Kruskal's Algorithm: Proof



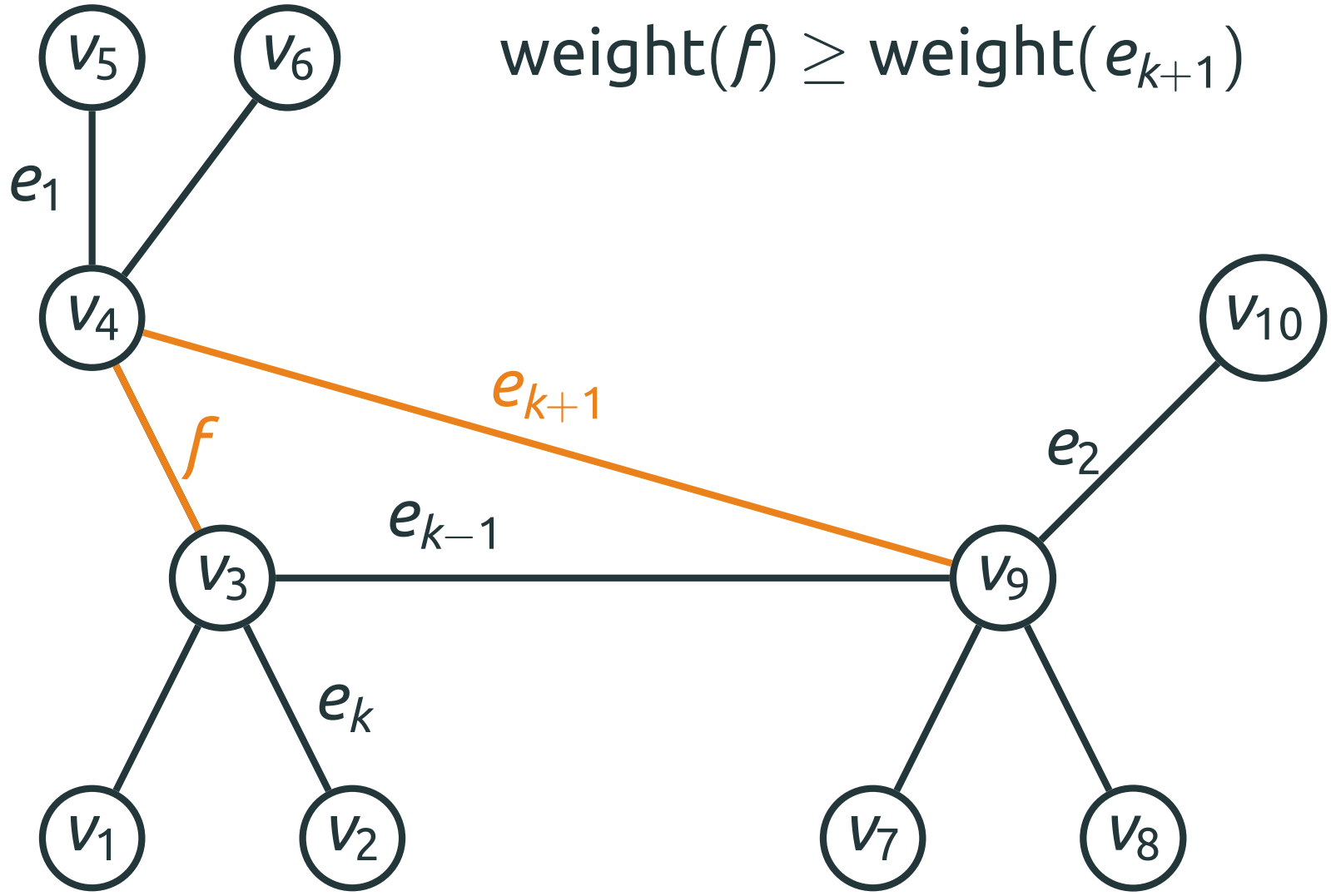
Kruskal's Algorithm: Proof



Kruskal's Algorithm: Proof

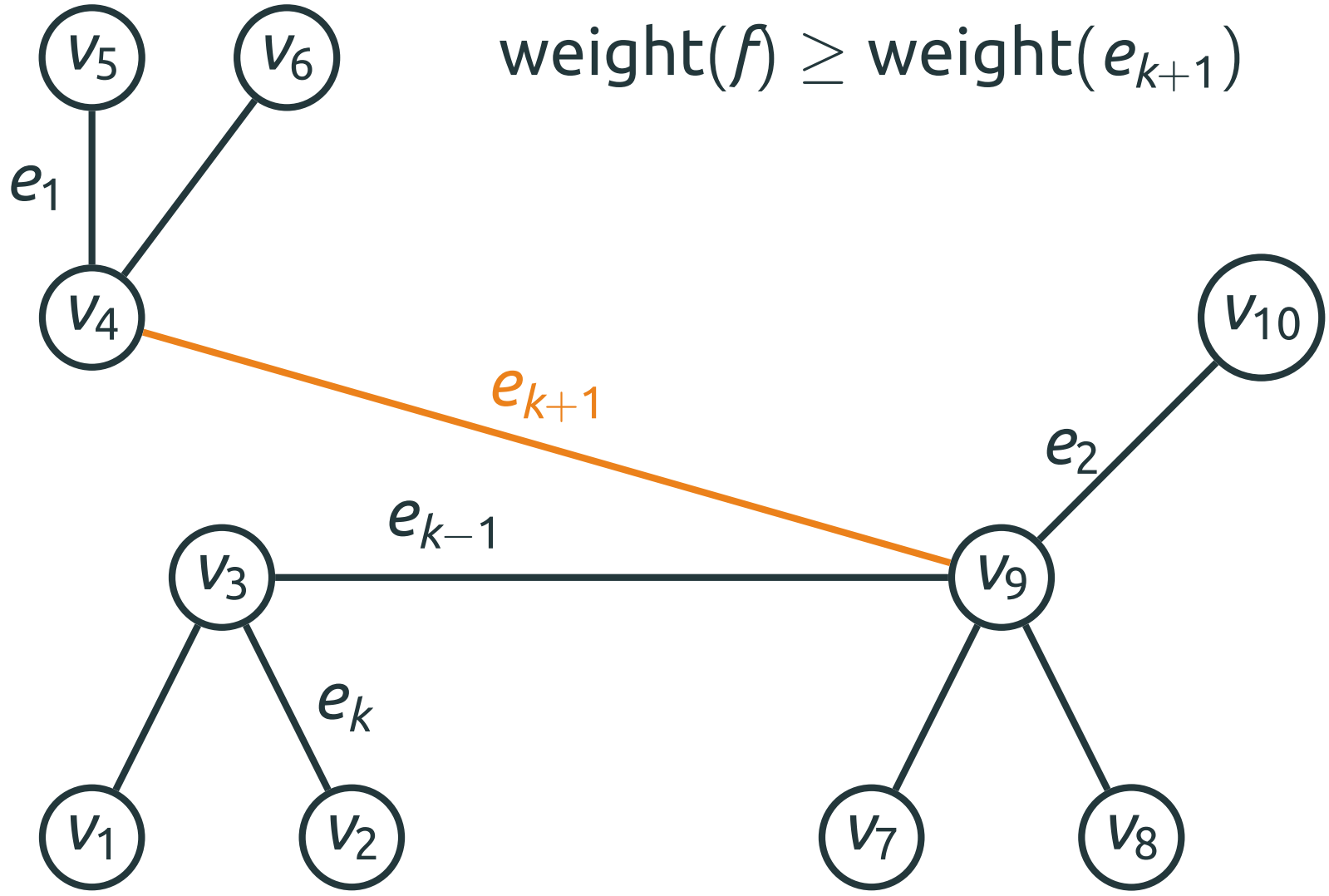


Kruskal's Algorithm: Proof



$\text{weight}(f) \geq \text{weight}(e_{k+1})$

Kruskal's Algorithm: Proof



Kruskal's Algorithm: Proof

